

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

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Lesson 19: Economic Growth (Summarized Notes)

Economic Growth

Economic growth refers to the increase in a country's production of goods and services over time. It improves national income, employment, and living standards. Per capita income is commonly used to compare the economic development of countries. Higher per capita income generally indicates a higher standard of living. Developed countries have much higher per capita income than developing countries.

Per Capita Income of Selected Countries

Norway had the highest per capita income among the listed countries. The United States, Japan, Switzerland, Germany, and the United Kingdom were among the high-income nations. Pakistan's per capita income was **US \$470**, which was much lower than developed countries. Bangladesh, Nigeria, and India also had relatively low per capita incomes.

The Solow Growth Model

The Solow Growth Model was developed by **Robert Solow**, who received the Nobel Prize for his work on economic growth. It explains how capital, labor, and technology determine economic growth. The model is widely used for economic policy analysis and as the foundation for modern growth theories. Economic growth depends on capital accumulation, labor force growth, and technological progress.

Assumptions of the Solow Model

Capital (K) is variable because investment increases it while depreciation decreases it. Labor (L) increases because of population growth. Government spending (G) and taxes (T) are ignored to simplify analysis. The consumption function is simplified. Technology is assumed to improve productivity.

Production Function

The production function shows how labor, capital, and technology are transformed into output.

Aggregate production function:

$$Y = F(K, L)$$

Output per worker is:

$$y = Y/L$$

Capital per worker is:

$$k = K/L$$

Under constant returns to scale:

$$y = f(k)$$

Output per worker depends only on capital per worker.

National Income Identity

Without government:

$$Y = C + I$$

Per worker form:

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$$y = c + i$$

Total output is divided between consumption and investment.

Consumption Function

Saving rate is represented by s .

Consumption per worker:

$$c = (1 - s)y$$

Higher saving means lower current consumption.

Saving and Investment

Saving per worker:

$$\text{Saving} = sy$$

Investment equals saving.

Investment function:

$$i = sy = sf(k)$$

More saving increases investment and future capital formation.

Output, Consumption and Investment

Output per worker increases with capital per worker.

Investment is a fraction of output determined by the saving rate.

Consumption equals the remaining income after saving.

The production function shows diminishing marginal product of capital (MPK).

Depreciation

Depreciation is represented by δ .

It is the fraction of capital stock that wears out every period.

Depreciation per worker:

$$\delta k$$

Higher depreciation reduces capital accumulation.

Capital Accumulation

Investment increases capital stock.

Depreciation decreases capital stock.

Net change in capital stock:

$$\Delta k = \text{Investment} - \text{Depreciation}$$

Substituting investment:

$$\Delta k = sf(k) - \delta k$$

Equation of Motion for Capital

The central equation of the Solow Model is:

$$\Delta k = sf(k) - \delta k$$

It determines changes in capital over time.

Income, output, and consumption all depend on capital per worker.

Steady State

Steady state occurs when investment exactly equals depreciation.

At steady state:

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$$sf(k) = \delta k$$

There is no change in capital stock.

Therefore:

$$\Delta k = 0$$

Steady-state capital is represented by k^* .

Important Exam Points

Robert Solow won the Nobel Prize for contributions to economic growth theory.

The Solow Growth Model explains growth through capital, labor, and technology.

Production Function: $Y = F(K, L)$.

Output per worker: $y = Y/L = f(k)$.

Capital per worker: $k = K/L$.

National Income Identity: $Y = C + I$.

Per worker identity: $y = c + i$.

Consumption Function: $c = (1 - s)y$.

Saving equals Investment: $i = sy = sf(k)$.

Depreciation per worker is δk .

Capital accumulation equation: $\Delta k = sf(k) - \delta k$.

Steady State occurs when $sf(k) = \delta k$ and $\Delta k = 0$.

The production function exhibits **Diminishing Marginal Product of Capital (MPK)**.

Higher saving increases investment and future capital accumulation.

Pakistan's per capita income (2004) was **US \$470**.

Important MCQs

1. Who developed the Solow Growth Model?

- A) John Maynard Keynes
- B) Milton Friedman
- C) Robert Solow ✓**
- D) David Ricardo

2. Which equation represents the production function?

- A) $Y = C + I$
- B) $Y = F(K, L)$ ✓**
- C) $C = (1 - s)Y$
- D) $\Delta K = I + D$

3. In the Solow Model, steady state occurs when:

- A) Saving is zero
- B) Consumption equals income
- C) Investment equals depreciation ✓**
- D) Capital equals labor

4. Which equation represents the consumption function?

- A) $c = sy$
- B) $c = (1 - s)y$ ✓**

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- C) $c = y + i$
D) $c = \delta k$

5. The central equation of the Solow Model is:

- A) $Y = C + I$
B) $y = f(k)$
C) $\Delta k = sf(k) - \delta k$ ✓
D) $MPK = Y/K$

Short Conceptual Questions

Q1. Why is the Solow Growth Model important?

Answer:

It explains how capital, labor, and technology influence long-run economic growth. It is widely used in economic policy and growth analysis.

Q2. What is the steady state in the Solow Model?

Answer:

The steady state is the point where investment exactly equals depreciation. At this stage, capital per worker remains constant over time.

Numerical / Formula Questions

Question 1

If the saving rate $s = 0.25$ and output per worker $y = 120$, calculate investment per worker.

Solution

Investment = sy

= 0.25×120

= 30

Answer: Investment per worker = 30

Question 2

If $sf(k) = 45$ and $\delta k = 30$, calculate the change in capital per worker (Δk).

Solution

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$$\Delta k = sf(k) - \delta k$$

$$= 45 - 30$$

$$= 15$$

Answer: $\Delta k = 15$ units (capital stock increases by 15).

Lesson 20: Solow Growth Model – Steady State and Saving (Summarized Notes)

Capital Movement Toward the Steady State

When capital per worker is below the steady-state level ($k < k^*$), investment is greater than depreciation.

As a result, capital stock continues to increase over time.

When capital per worker is above the steady state ($k > k^*$), depreciation exceeds investment.

This causes the capital stock to decrease until it reaches the steady state.

The economy naturally moves toward the steady-state level of capital.

Steady State Equilibrium

The steady state is the point where investment exactly equals depreciation.

At this point, there is no change in capital per worker.

The condition for steady state is:

$$sf(k) = \delta k^{**}$$

Since investment equals depreciation:

$$\Delta k = 0$$

Capital, output, and consumption remain constant unless saving or technology changes.

Numerical Example of the Solow Model

Aggregate production function:

$$Y = K^{\frac{1}{3}} L^{\frac{2}{3}}$$

Per-worker production function:

$$y = k^{\frac{1}{3}}$$

Assumptions:

Saving rate $s = 0.3$

Depreciation rate $\delta = 0.1$

Initial capital per worker $k = 4$

As time passes, investment increases capital while depreciation reduces it.

Capital gradually approaches the steady-state value.

Approaching the Steady State

Initially, investment exceeds depreciation, so capital increases.

As capital increases, depreciation also increases.

The gap between investment and depreciation becomes smaller.

Eventually, investment equals depreciation.

At this point, the economy reaches the steady state where capital no longer changes.

Solving for the Steady State

At steady state:

$$\Delta k = 0$$

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Using the equation:

$$sf(k) = \delta k^{**}$$

With:

$$s = 0.3$$

$$\delta = 0.1$$

$$y = k^{\frac{1}{2}}$$

The steady-state values are:

$$k = 9^*$$

$$y = 3^*$$

Consumption at steady state:

$$c = (1 - s)y^{**}$$

$$c = 2.1^*$$

Increase in the Saving Rate

An increase in the saving rate raises investment.

Higher investment increases the capital stock.

The economy moves toward a new, higher steady-state level.

A higher steady-state capital stock results in higher output per worker.

Countries with higher saving and investment rates achieve higher long-run income levels.

International Evidence

Countries with higher investment rates generally have higher income per person.

Developed countries usually save and invest a larger share of their output.

Low-income countries often have lower investment rates and lower per capita income.

The Solow Model supports the positive relationship between saving, investment, and long-run economic growth.

Important Exam Points

Robert Solow explained long-run growth through capital accumulation.

When $k < k^*$, investment exceeds depreciation and capital increases.

When $k > k^*$, depreciation exceeds investment and capital decreases.

Steady-state condition is $sf(k) = \delta k^{**}$.

At steady state, $\Delta k = 0$.

Aggregate production function: $Y = K^{\frac{1}{2}}L^{\frac{1}{2}}$.

Per-worker production function: $y = k^{\frac{1}{2}}$.

Steady-state capital in the numerical example is $k = 9^*$.

Steady-state output is $y = 3^*$.

Steady-state consumption is $c = 2.1^*$.

Higher saving increases investment and long-run capital stock.

Higher saving leads to higher output and income per worker.

Countries with higher investment rates generally have higher per capita income.

Important MCQs

1. When capital per worker is below the steady state, what happens?

- A) Depreciation exceeds investment
- B) Investment exceeds depreciation ✓**
- C) Capital remains constant
- D) Output decreases immediately

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2. What is the steady-state condition in the Solow Model?

- A) Saving equals consumption
- B) Output equals consumption
- C) Investment equals depreciation ✓
- D) Labor equals capital

3. In the numerical example, the steady-state capital per worker is:

- A) 3
- B) 6
- C) 9 ✓
- D) 12

4. The per-worker production function in this lesson is:

- A) $y = k$
- B) $y = k^2$
- C) $y = k^{1/2}$ ✓
- D) $y = 1/k$

5. According to the Solow Model, an increase in the saving rate will:

- A) Reduce investment
- B) Lower output
- C) Increase capital and income per worker ✓
- D) Eliminate depreciation

Short Conceptual Questions

Q1. Why does the economy move toward the steady state?

Answer:

When investment and depreciation are unequal, capital either rises or falls until they become equal. This adjustment naturally brings the economy to the steady-state equilibrium.

Q2. How does a higher saving rate affect economic growth?

Answer:

A higher saving rate increases investment and capital accumulation. This raises long-run output and income per worker.

Numerical / Formula Questions

Question 1

Given $s = 0.3$, $\delta = 0.1$, and $k = 9$, calculate y and c .

Solution

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Production:

$$y = \sqrt{k} = \sqrt{9} = 3$$

Consumption:

$$c = (1 - s)y$$

$$= 0.7 \times 3$$

$$= 2.1$$

Answer: $y = 3$, $c = 2.1$

Question 2

If $sf(k) = 1.2$ and $\delta k = 0.9$, calculate the change in capital per worker (Δk).

Solution

$$\Delta k = sf(k) - \delta k$$

$$= 1.2 - 0.9$$

$$= 0.3$$

Answer: $\Delta k = 0.3$ (capital stock increases by 0.3 units).

Lesson 21: Economic Growth (Continued) – Golden Rule and Population Growth (Summarized Notes)

The Golden Rule

Different saving rates lead to different steady-state levels of capital and consumption.

The best steady state is the one that provides the **highest consumption per worker**.

Economic well-being depends mainly on consumption rather than production alone.

The Golden Rule identifies the saving rate and capital stock that maximize long-run consumption.

Golden Rule Level of Capital Stock

The Golden Rule level of capital is represented by **k^* gold**.

It is the steady-state capital stock that gives the maximum consumption per worker.

Steady-state consumption is:

$$c = f(k) - \delta k^{***}$$

Since investment equals depreciation at steady state:

$$i = \delta k^{**}$$

The Golden Rule occurs where the gap between output and depreciation is the greatest.

Golden Rule Condition

The Golden Rule is achieved when:

$$MPK = \delta$$

Where:

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MPK = Marginal Product of Capital

δ = Depreciation Rate

At this point, consumption reaches its maximum possible level.

Transition to the Golden Rule Steady State

The economy does not automatically move toward the Golden Rule.

Government policies must adjust the saving rate to reach it.

Changing the saving rate creates a new steady state with higher long-run consumption.

Starting with Too Much Capital

If the economy has more capital than the Golden Rule level ($k > k_{gold}^{**}$), the saving rate should be reduced.

Lower saving increases present consumption.

Consumption becomes higher throughout the adjustment period.

Starting with Too Little Capital

If the economy has less capital than the Golden Rule level ($k < k_{gold}^{**}$), the saving rate should be increased.

Current consumption falls because more income is saved.

Future generations benefit from higher consumption after reaching the new steady state.

Need for Population Growth and Technology

The basic Solow Model cannot explain permanent economic growth.

High saving increases growth only temporarily.

Sustained economic growth requires:

Population growth.

Technological progress.

Population Growth

Population and labor force grow at an exogenous rate n .

Population growth increases the number of workers in the economy.

More workers require additional capital to maintain productivity.

Population growth formula:

$$\Delta L/L = n$$

Break-Even Investment

Break-even investment is the investment required to keep capital per worker constant.

It includes:

Replacement of worn-out capital (δk).

Capital for new workers (nk).

Formula:

$$\text{Break-even Investment} = (\delta + n)k$$

Equation of Motion with Population Growth

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When population grows, the equation becomes:

$$\Delta k = sf(k) - (\delta + n)k$$

Investment must now cover both depreciation and population growth.

If investment exceeds break-even investment, capital per worker rises.

Impact of Population Growth

Higher population growth raises break-even investment.

More investment is needed just to maintain existing capital per worker.

Higher population growth reduces steady-state capital.

Lower capital per worker leads to lower income per worker.

Countries with rapid population growth generally have lower long-run income levels.

Golden Rule with Population Growth

Steady-state consumption becomes:

$$c = f(k) - (\delta + n)k^{***}$$

Consumption is maximized when:

$$MPK = \delta + n$$

Or equivalently:

$$MPK - \delta = n$$

At the Golden Rule with population growth, the net marginal product of capital equals the population growth rate.

Important Exam Points

The Golden Rule identifies the steady state with maximum consumption per worker.

Golden Rule capital stock is denoted by **k^{*gold}** .

Steady-state consumption formula: $c = f(k) - \delta k^{***}$.

At steady state: $i = \delta k^{**}$.

Golden Rule condition: **$MPK = \delta$** .

The economy does not automatically move to the Golden Rule.

If $k > k^{gold**}$, reduce the saving rate.

If $k < k^{gold**}$, increase the saving rate.

The basic Solow Model cannot explain sustained long-run growth.

Permanent growth requires population growth and technological progress.

Population growth rate is represented by **n** .

Break-even investment formula: **$(\delta + n)k$** .

Equation of motion with population growth: $\Delta k = sf(k) - (\delta + n)k$.

Higher population growth lowers steady-state capital and income per worker.

Golden Rule with population growth: **$MPK = \delta + n$** .

Equivalent condition: **$MPK - \delta = n$** .

Important MCQs

1. What is the main objective of the Golden Rule?

- A) Maximize investment
- B) Maximize saving
- C) **Maximize consumption per worker ✓**
- D) Maximize population growth

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2. At the Golden Rule level of capital, the condition is:

- A) $MPK = n$
- B) $MPK = 0$
- C) $MPK = \delta$ ✓
- D) $MPK = s$

3. With population growth, the equation of motion becomes:

- A) $\Delta k = sf(k) - \delta k$
- B) $\Delta k = sf(k) + nk$
- C) $\Delta k = sf(k) - (\delta + n)k$ ✓
- D) $\Delta k = y - c$

4. Break-even investment includes:

- A) Only depreciation
- B) Only new workers' capital
- C) Depreciation and capital for new workers ✓
- D) Consumption and investment

5. According to the Solow Model, higher population growth results in:

- A) Higher steady-state income
- B) Higher capital per worker
- C) Lower capital and income per worker ✓
- D) No effect on growth

Short Conceptual Questions

Q1. Why does the economy not automatically reach the Golden Rule?

Answer:

The saving rate is determined by policy and does not adjust automatically. Government intervention is needed to achieve the Golden Rule steady state.

Q2. Why does population growth reduce capital per worker?

Answer:

More workers require additional capital to maintain the same capital-worker ratio. If investment is insufficient, capital per worker declines.

Numerical / Formula Questions

Question 1

If $\delta = 0.05$, $n = 0.03$, and $k = 200$, calculate break-even investment.

Solution

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Break-even Investment = $(\delta + n)k$

$$= (0.05 + 0.03) \times 200$$

$$= 0.08 \times 200$$

$$= 16$$

Answer: Break-even investment = 16 units.

Question 2

If $sf(k) = 30$, $\delta = 0.04$, $n = 0.02$, and $k = 400$, calculate Δk .

Solution

Break-even Investment = $(\delta + n)k$

$$= (0.04 + 0.02) \times 400$$

$$= 0.06 \times 400$$

$$= 24$$

$$\Delta k = sf(k) - (\delta + n)k$$

$$= 30 - 24$$

$$= 6$$

Answer: $\Delta k = 6$ units (capital per worker increases by 6).

Lesson 22: Economic Growth (Continued) – Technological Progress in the Solow Model (Summarized Notes)

Technological Progress in the Solow Model

The basic Solow Model assumes technology remains constant.

In reality, technology continuously improves and increases productivity.

Technological progress is an important source of sustained long-run economic growth.

Unlike the basic Solow Model, income per capita can continue to grow because of technological improvements.

Labor Efficiency

Labor efficiency is represented by E .

Technological progress increases labor efficiency at an exogenous growth rate g .

Formula:

$$g = \Delta E / E$$

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Higher labor efficiency allows workers to produce more output with the same amount of labor.

Production Function with Technological Progress

The production function becomes:

$$Y = F(K, LE)$$

Where:

$$L \times E = \text{Effective Workers}$$

An increase in labor efficiency has the same effect as an increase in the labor force.

Output and Capital per Effective Worker

Output per effective worker:

$$y = Y/LE$$

Capital per effective worker:

$$k = K/LE$$

Production function:

$$y = f(k)$$

Saving and investment per effective worker:

$$sy = sf(k)$$

Economic performance now depends on capital per effective worker rather than capital per worker.

Break-Even Investment with Technological Progress

To keep capital per effective worker constant, investment must cover:

Replacement of depreciated capital (δk).

Capital for new workers (nk).

Capital for more efficient workers created by technology (gk).

Formula:

$$\text{Break-even Investment} = (\delta + n + g)k$$

Equation of Motion

With technological progress:

$$\Delta k = sf(k) - (\delta + n + g)k$$

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If investment exceeds break-even investment, capital per effective worker increases.
If investment equals break-even investment, the economy reaches steady state.

Steady-State Growth Rates

Capital per effective worker (**k**) remains constant in steady state.
Output per effective worker (**y**) also remains constant.
Output per worker grows at the technological progress rate (**g**).
Total output grows at the combined rate of population growth and technological progress (**n + g**).

Golden Rule with Technological Progress

Steady-state consumption becomes:

$$c = f(k) - (\delta + n + g)k^{***}$$

Consumption is maximized when:

$$MPK = \delta + n + g$$

Equivalent condition:

$$MPK - \delta = n + g$$

At the Golden Rule, the net marginal product of capital equals population growth plus technological progress.

Policies to Promote Economic Growth

Governments should determine whether the saving rate is appropriate.
Policies can encourage higher saving and investment.
Investment should be balanced among:
Private physical capital.
Public infrastructure.
Human capital such as education and training.
Governments should also encourage research, innovation, and technological progress.

Evaluating the Saving Rate

The Golden Rule helps determine whether saving is too high or too low.

If:

$$MPK - \delta > n + g$$

The economy is below the Golden Rule.
The saving rate should be increased.

If:

$$MPK - \delta < n + g$$

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The economy is above the Golden Rule.
The saving rate should be reduced.

Numerical Evaluation

Given:

$$k = 2.5y$$

$$\delta k = 0.1y$$

$$MPK \times k = 0.3y$$

Depreciation rate:

$$\delta = 0.1 / 2.5 = 0.04$$

Marginal Product of Capital:

$$MPK = 0.3 / 2.5 = 0.12$$

Net Marginal Product of Capital:

$$MPK - \delta = 0.12 - 0.04 = 0.08$$

Since:

$$0.08 > 0.03 (n + g)$$

The economy is below the Golden Rule.
Increasing the saving rate will increase long-run consumption and economic growth.

Important Exam Points

Technological progress is represented by E .
Growth rate of labor efficiency is $g = \Delta E/E$.
Production function: $Y = F(K, LE)$.
Effective workers = $L \times E$.
Output per effective worker: $y = Y/LE$.
Capital per effective worker: $k = K/LE$.
Break-even investment: $(\delta + n + g)k$.
Equation of motion: $\Delta k = sf(k) - (\delta + n + g)k$.
Output per worker grows at rate g .
Total output grows at rate $n + g$.
Golden Rule condition: $MPK = \delta + n + g$.
Equivalent condition: $MPK - \delta = n + g$.
If $MPK - \delta > n + g$, increase the saving rate.
If $MPK - \delta < n + g$, decrease the saving rate.
Calculated values:
 $\delta = 0.04$
 $MPK = 0.12$
 $MPK - \delta = 0.08$

Since $0.08 > 0.03$, the economy is below the Golden Rule.

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Important MCQs

1. In the Solow Model, labor efficiency is represented by:

- A) K
- B) L
- C) E ✓
- D) Y

2. The production function with technological progress is:

- A) $Y = C + I$
- B) $Y = F(K, L)$
- C) $Y = F(K, LE)$ ✓
- D) $Y = K + L$

3. Break-even investment with technological progress is:

- A) δk
- B) $(\delta + n)k$
- C) $(\delta + n + g)k$ ✓
- D) $sf(k)$

4. The Golden Rule with technological progress is achieved when:

- A) $MPK = \delta$
- B) $MPK = n$
- C) $MPK = \delta + n + g$ ✓
- D) $MPK = g$

5. If $MPK - \delta$ is greater than $(n + g)$, the economy should:

- A) Reduce investment
- B) Reduce saving
- C) Increase the saving rate ✓
- D) Reduce technology

Short Conceptual Questions

Q1. Why is technological progress important in the Solow Model?

Answer:

Technological progress increases labor productivity and allows output per worker to grow continuously. It explains sustained long-run economic growth.

Q2. What is meant by effective workers?

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Answer:

Effective workers are measured as labor multiplied by labor efficiency (LE). They represent workers whose productivity is enhanced through technology.

Numerical / Formula Questions

Question 1

If $\delta = 0.05$, $n = 0.02$, $g = 0.03$, and $k = 150$, calculate break-even investment.

Solution

Break-even Investment = $(\delta + n + g)k$

$$= (0.05 + 0.02 + 0.03) \times 150$$

$$= 0.10 \times 150$$

$$= 15$$

Answer: Break-even investment = 15 units.

Question 2

Given:

$$k = 2.5y$$

$$\delta k = 0.1y$$

$$MPK \times k = 0.3y$$

Calculate δ , MPK, and $MPK - \delta$.

Solution

$$\delta = 0.1 \div 2.5 = 0.04$$

$$MPK = 0.3 \div 2.5 = 0.12$$

$$MPK - \delta = 0.12 - 0.04 = 0.08$$

Answer: $\delta = 0.04$, $MPK = 0.12$, $MPK - \delta = 0.08$.

Lesson 23: Economic Growth (Continued) – Growth Policies and Endogenous Growth Theory (Summarized Notes)

Policies to Increase the Saving Rate

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Governments can promote economic growth by increasing national saving.
Reducing the government budget deficit or increasing the budget surplus raises national saving.
Private saving can be encouraged through tax reforms.
Lower taxes on capital gains, corporate income, and estates increase incentives to save.
Replacing income tax with a consumption tax also encourages saving.
Tax incentives for retirement accounts increase long-term savings.

Allocating the Economy's Investment

Investment can be divided into three major categories:
Private capital.
Public infrastructure.
Human capital through education and skills.
Proper allocation of investment improves long-run economic growth.

Views on Investment Allocation

One view suggests equal tax treatment for all industries so markets allocate investment efficiently.
The second view supports industrial policy where the government promotes industries with large social benefits.
Industrial policy aims to encourage industries that create positive externalities.

Problems with Industrial Policy

Governments may fail to identify industries with the highest future returns.
Political interests may influence investment decisions instead of economic efficiency.
Poor policy choices may lead to inefficient allocation of resources.

Encouraging Technological Progress

Governments encourage innovation through patent laws.
Patents provide temporary monopoly rights that reward inventors.
Tax incentives increase private research and development (R&D).
Research grants support universities and scientific research.
Industrial policy may also promote high-technology industries.

Growth Empirics

The Solow Model predicts balanced economic growth.
Output per worker and capital per worker grow at the same rate.
The capital-output ratio remains approximately constant over time.
Real wages rise with productivity, while the real rental price of capital remains relatively stable.
These predictions closely match real-world evidence.

Convergence

The Solow Model predicts that poorer countries should grow faster than richer countries if other conditions are similar.
This process is called convergence.
In reality, countries differ in saving rates, population growth, and education.
Therefore, countries converge only toward their own steady states.
This concept is known as conditional convergence.

Factor Accumulation and Production Efficiency

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Differences in income per capita result from:
Differences in physical and human capital.
Differences in production efficiency.
Countries with greater capital usually have higher production efficiency.
Capital accumulation and efficiency reinforce each other.

Endogenous Growth Theory

Unlike the Solow Model, technological progress is generated within the economy.
Growth depends on investment, innovation, and knowledge creation.
The production function is:

$$Y = AK$$

Where:

A = Output produced by each unit of capital

The marginal product of capital remains constant.
Investment becomes the main engine of long-run economic growth.

Equation of Motion in Endogenous Growth

Investment:

$$sY$$

Depreciation:

$$\delta K$$

Capital accumulation:

$$\Delta K = sY - \delta K$$

Growth rate:

$$\Delta Y/Y = \Delta K/K = sA - \delta$$

If:

$$sA > \delta$$

Income grows continuously without reaching a steady state.

Returns to Capital

The Solow Model assumes diminishing returns to physical capital.
Endogenous Growth Theory assumes broader capital, including knowledge and human capital, may exhibit constant or increasing returns.
Knowledge generates spillover benefits that support continuous growth.

Two-Sector Endogenous Growth Model

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The economy contains:

Manufacturing firms producing goods.

Research institutions producing knowledge.

A fraction of labor (u) is employed in research.

Research increases labor efficiency.

Higher research effort raises long-run growth.

Effects of Saving and Research

Saving rate (s) increases the level of income but not the long-run growth rate.

Research effort (u) affects both income level and long-run growth.

More resources devoted to research can increase technological progress.

Research and Development (R&D)

Private firms conduct research to earn profits.

Patents reward innovation by providing temporary monopoly rights.

Innovation creates positive externalities that benefit future research.

The social return to R&D is estimated to exceed 40% annually.

Governments often encourage R&D because private firms invest less than the socially optimal amount.

Important Exam Points

Increasing national saving promotes long-run economic growth.

Budget surpluses increase national saving.

Investment categories are private capital, public infrastructure, and human capital.

Industrial policy supports industries with positive externalities.

Patent laws encourage innovation.

Tax incentives increase research and development.

The Solow Model predicts balanced growth.

Conditional convergence means countries converge to their own steady states.

Income differences arise from capital accumulation and production efficiency.

Endogenous Growth Theory explains technological progress internally.

Production function: $Y = AK$.

Capital accumulation equation: $\Delta K = sY - \delta K$.

Growth equation: $\Delta Y/Y = sA - \delta$.

If $sA > \delta$, economic growth continues indefinitely.

The Solow Model assumes diminishing returns to capital.

Endogenous Growth Theory assumes constant or increasing returns to broader capital.

Research effort (u) affects long-run growth.

Positive externalities justify government support for R&D.

Estimated social return to R&D is **at least 40% per year**.

Important MCQs

1. Which policy can increase the national saving rate?

A) Increasing income taxes only

B) Increasing imports

C) Reducing the government budget deficit ✓

D) Increasing government consumption

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2. According to Endogenous Growth Theory, the production function is:

- A) $Y = F(K, L)$
- B) $Y = C + I$
- C) $Y = AK$ ✓
- D) $Y = K^2$

3. Conditional convergence means countries:

- A) Reach the same income immediately
- B) Never converge
- C) Converge toward their own steady states ✓
- D) Grow at identical rates forever

4. In the endogenous growth model, permanent growth occurs when:

- A) $sA < \delta$
- B) $sA = \delta$
- C) $sA > \delta$ ✓
- D) $MPK = 0$

5. Which factor directly affects both the level and growth rate of income in the two-sector model?

- A) Saving rate (s)
- B) Depreciation rate (δ)
- C) Research labor (u) ✓
- D) Population growth (n)

Short Conceptual Questions

Q1. Why does the government encourage Research and Development (R&D)?

Answer:

R&D creates positive externalities that benefit the whole economy beyond private firms. Government support increases innovation and long-run economic growth.

Q2. What is conditional convergence?

Answer:

Conditional convergence means countries move toward their own steady-state income levels. These steady states depend on saving, education, and population growth.

Numerical / Formula Questions

Question 1

If $s = 0.30$, $A = 5$, and $\delta = 0.80$, calculate the economic growth rate using the endogenous growth model.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Solution

Growth Rate:

$$\Delta Y/Y = sA - \delta$$

$$= 0.30 \times 5 - 0.80$$

$$= 1.50 - 0.80$$

$$= 0.70$$

Answer: Economic growth rate = 0.70 (70%).

Question 2

If $Y = 800$, $s = 0.25$, $\delta K = 120$, calculate capital accumulation (ΔK).

Solution

Investment:

$$sY = 0.25 \times 800 = 200$$

Capital Accumulation:

$$\Delta K = sY - \delta K$$

$$= 200 - 120$$

$$= 80$$

Answer: $\Delta K = 80$ units.

Lesson 24: Aggregate Demand and Aggregate Supply (Summarized Notes)

Time Horizons

Macroeconomics studies the economy in both the short run and the long run.

In the **long run**, prices are flexible and adjust to changes in supply and demand.

In the **short run**, many prices remain sticky or fixed.

Because prices are sticky, the economy behaves differently in the short run than in the long run.

Classical Macroeconomic Theory

Classical theory assumes prices are completely flexible.

Output is determined by the supply side of the economy.

The main determinants of output are capital, labor, and technology.

Changes in aggregate demand affect only the price level, not real output.

Classical theory mainly explains the long-run behavior of the economy.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

When Prices are Sticky

When prices do not adjust quickly, output and employment depend on aggregate demand.

Aggregate demand is influenced by:

Fiscal policy (Government Spending **G** and Taxes **T**).

Monetary policy (Money Supply **M**).

Changes in Consumption (**C**) and Investment (**I**).

Demand fluctuations directly affect production and employment in the short run.

Aggregate Demand and Aggregate Supply Model

The AD-AS model explains fluctuations in output, employment, and prices.

It is widely used by economists and policymakers.

The model shows how equilibrium output and the price level are determined in both the short run and long run.

Aggregate Demand (AD)

The Aggregate Demand curve shows the relationship between the price level and total output demanded.

It is based on the Quantity Theory of Money.

Quantity Equation:

$$MV = PY$$

Money Demand Function:

$$(M/P)^d = kY$$

For a given money supply and velocity, the price level and output are inversely related.

Downward-Sloping Aggregate Demand Curve

An increase in the price level reduces real money balances (**M/P**).

Lower real money balances reduce spending.

As a result, aggregate demand decreases.

Therefore, the AD curve slopes downward.

Shifting the Aggregate Demand Curve

An increase in the money supply shifts the AD curve to the right.

A decrease in the money supply shifts the AD curve to the left.

Monetary policy is a major factor causing shifts in aggregate demand.

Long-Run Aggregate Supply (LRAS)

Long-run output depends only on productive resources and technology.

Production Function:

$$Y = F(K, L)$$

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Full-employment output is called the natural level of output.
At full employment, unemployment equals the natural rate.
Since output does not depend on the price level, the LRAS curve is vertical.

Long-Run Effect of an Increase in Money Supply

An increase in the money supply shifts AD to the right.
In the long run, output remains unchanged.
Only the price level increases.
Money is neutral in the long run.

Short-Run Aggregate Supply (SRAS)

In the short run, prices are sticky.
Firms are willing to supply whatever quantity consumers demand at the fixed price.
Therefore, the SRAS curve is horizontal.
Output changes while the price level remains fixed.

Short-Run Effect of an Increase in Money Supply

An increase in the money supply raises aggregate demand.
Since prices are fixed, firms increase production instead of prices.
Output and employment rise.
The price level remains unchanged in the short run.

Adjustment from Short Run to Long Run

Over time, prices gradually become flexible.
If actual output is greater than full-employment output ($Y > \bar{Y}$), prices rise.
If actual output is less than full-employment output ($Y < \bar{Y}$), prices fall.
If output equals full-employment output ($Y = \bar{Y}$), prices remain stable.
This price adjustment moves the economy toward long-run equilibrium.

Short-Run and Long-Run Effects of Monetary Expansion

Initially, the economy is at long-run equilibrium.
An increase in money supply shifts AD to the right.
In the short run, output rises while prices remain fixed.
As prices gradually adjust upward, SRAS shifts.
Eventually, the economy returns to full-employment output with a higher price level.
The final long-run equilibrium has higher prices but the same level of output.

Important Exam Points

Long run assumes flexible prices.
Short run assumes sticky prices.
Classical theory explains long-run output.
Output depends on capital, labor, and technology.
Aggregate Demand equation: $MV = PY$.
Money demand equation: $(M/P)d = kY$.
The AD curve slopes downward because higher prices reduce real money balances.
An increase in money supply shifts AD to the right.
Long-run production function: $Y = F(K, L)$.
LRAS curve is vertical because output is independent of the price level.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

SRAS curve is horizontal because prices are sticky.

In the short run, higher money supply increases output and employment.

In the long run, higher money supply only increases the price level.

Money is neutral in the long run.

If $Y > \bar{Y}$, prices rise.

If $Y < \bar{Y}$, prices fall.

If $Y = \bar{Y}$, prices remain constant.

The economy eventually returns to full-employment output after price adjustment.

Important MCQs

1. In the long run, output is mainly determined by:

- A) Aggregate demand
- B) Government spending
- C) **Capital, labor, and technology ✓**
- D) Money supply

2. Which equation represents the Quantity Theory of Money?

- A) $Y = C + I$
- B) $Y = F(K, L)$
- C) **$MV = PY$ ✓**
- D) $\Delta K = I - \delta K$

3. Why does the Aggregate Demand curve slope downward?

- A) Output decreases automatically
- B) Prices increase production
- C) **Higher prices reduce real money balances ✓**
- D) Taxes increase demand

4. In the short run, the SRAS curve is:

- A) Vertical
- B) Downward sloping
- C) **Horizontal ✓**
- D) Upward sloping

5. In the long run, an increase in the money supply mainly causes:

- A) Higher output
- B) Lower employment
- C) **Higher price level ✓**
- D) Lower production

Short Conceptual Questions

Q1. Why is the LRAS curve vertical?

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Answer:

Long-run output depends only on capital, labor, and technology, not on prices.
Therefore, changes in the price level do not affect full-employment output.

Q2. Why is the SRAS curve horizontal?

Answer:

Prices are assumed to be sticky in the short run.
Firms supply whatever quantity consumers demand at the fixed price level.

Numerical / Formula Questions

Question 1

Using the Quantity Equation $MV = PY$, calculate the price level (P) if:

$$M = 800, V = 4, Y = 1600$$

Solution

$$[P = \frac{MV}{Y}]$$

$$[P = \frac{800 \times 4}{1600}]$$

$$[P = \frac{3200}{1600} = 2]$$

Answer: Price Level (P) = 2

Question 2

Using $MV = PY$, calculate **Output (Y)** if:

$$M = 500, V = 6, P = 3$$

Solution

$$[Y = \frac{MV}{P}]$$

$$[Y = \frac{500 \times 6}{3}]$$

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$$Y = \frac{3000}{3} = 1000$$

Answer: Output (Y) = 1000 units.

Lesson 25: Aggregate Demand and Aggregate Supply (Continued) – Demand and Supply Shocks (Summarized Notes)

Shocks in the Economy

Shocks are unexpected (exogenous) changes in aggregate demand or aggregate supply. They temporarily move the economy away from full-employment equilibrium. Shocks may be either demand shocks or supply shocks. The economy gradually returns to long-run equilibrium as prices adjust.

Demand Shock

A demand shock is an unexpected change in aggregate demand. An increase in aggregate demand shifts the AD curve to the right. Output rises above its natural level in the short run. As prices gradually increase, output returns to its natural level. The economy moves from the short-run equilibrium back to the long-run equilibrium.

Negative Demand Shock

A decrease in the velocity of money reduces aggregate demand. If the money supply remains constant, lower velocity means less spending. The AD curve shifts to the left. Output and employment decrease in the short run. Over time, prices fall and the economy returns to full-employment output.

Supply Shocks

A supply shock changes production costs and affects firms' prices. Supply shocks are also known as price shocks. An adverse supply shock increases production costs and shifts the SRAS curve upward (left). A favorable supply shock lowers production costs and shifts SRAS downward (right).

Examples of Adverse Supply Shocks

Bad weather reduces agricultural production and increases food prices. Labor unions negotiate higher wages, increasing production costs. Environmental regulations increase firms' compliance costs. These events reduce output and raise the general price level.

Effects of an Adverse Supply Shock

The SRAS curve shifts upward. The economy experiences higher inflation. Output falls below the natural level. Employment decreases and unemployment rises. The economy gradually returns to equilibrium as prices adjust over time.

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Stabilization Policy

Stabilization policies reduce the severity of short-run economic fluctuations. Governments and central banks use fiscal and monetary policies to stabilize the economy. The central bank may increase aggregate demand through expansionary monetary policy. Such policies help restore output to full-employment. However, the long-run result is usually a permanently higher price level.

The Oil Shocks of the 1970s

During the early 1970s, OPEC reduced the global supply of oil. Oil prices increased sharply in 1973, 1974, and 1975. Higher oil prices increased production costs throughout the economy. The SRAS curve shifted upward. Inflation increased while output and employment declined. Unemployment also increased. As oil prices stabilized, the economy gradually recovered.

Late 1970s Oil Shock

While the economy was recovering, oil prices increased again. The second oil shock created another adverse supply shock. Inflation rose further. Output declined again. Unemployment increased once more.

The Oil Shocks of the 1980s

During the 1980s, oil prices fell significantly. Lower oil prices reduced production costs. The favorable supply shock shifted SRAS downward. Inflation decreased. Output increased. Unemployment declined as economic activity improved.

Important Exam Points

Shocks are unexpected changes in aggregate demand or aggregate supply. Demand shocks affect the AD curve. Supply shocks affect the SRAS curve. An increase in aggregate demand shifts AD to the right. A decrease in velocity shifts AD to the left. Negative demand shocks reduce output and employment. Supply shocks change production costs. Adverse supply shocks increase inflation and reduce output. Favorable supply shocks reduce prices and increase output. Examples of adverse supply shocks include bad weather, higher wages, and environmental regulations. Stabilization policy uses fiscal and monetary policy to reduce economic fluctuations. Expansionary monetary policy shifts AD to the right. Early 1970s OPEC oil crisis caused a major adverse supply shock. Effects of the 1970s oil shocks: Inflation increased. Output decreased. Unemployment increased. The 1980s fall in oil prices created a favorable supply shock.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Effects of the 1980s oil shock:
Inflation decreased.
Output increased.
Unemployment decreased.

Important MCQs

1. A demand shock primarily affects which curve?

- A) LRAS
- B) SRAS
- C) **Aggregate Demand (AD) ✓**
- D) Production Function

2. A decrease in the velocity of money causes:

- A) AD to shift right
- B) **AD to shift left ✓**
- C) SRAS to shift right
- D) LRAS to shift left

3. Which is an example of an adverse supply shock?

- A) Lower taxes
- B) Technological progress
- C) **Bad weather reducing crop yields ✓**
- D) Increase in productivity

4. The 1970s oil crisis mainly caused:

- A) Lower inflation
- B) Higher output
- C) **Higher inflation and lower output ✓**
- D) Lower unemployment

5. A favorable supply shock results in:

- A) Higher unemployment
 - B) Higher production costs
 - C) **Lower inflation and higher output ✓**
 - D) Lower investment
-

Short Conceptual Questions

Q1. What is a demand shock?

Answer:

A demand shock is an unexpected change in aggregate demand that affects output and employment. It shifts the Aggregate Demand (AD) curve either left or right.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Q2. Why did oil price increases create supply shocks in the 1970s?

Answer:

Higher oil prices increased production costs for firms across the economy.
This reduced output and increased inflation.

Numerical / Formula Questions

Question 1

Using the Quantity Equation $MV = PY$, calculate **Output (Y)** if:

M = 600, V = 5, P = 3

Solution

$$[Y = \frac{MV}{P}]$$

$$[Y = \frac{600 \times 5}{3}]$$

$$[Y = \frac{3000}{3} = 1000]$$

Answer: Output (Y) = 1000 units.

Question 2

Using $MV = PY$, calculate the **Price Level (P)** if:

M = 400, V = 6, Y = 1200

Solution

$$[P = \frac{MV}{Y}]$$

$$[P = \frac{400 \times 6}{1200}]$$

$$[P = \frac{2400}{1200} = 2]$$

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Answer: Price Level (P) = 2.

Lesson 26: Keynesian Theory of Income and Employment (Summarized Notes)

Keynesian Theory

The Keynesian theory explains income and employment in the short run.

In the **long run**, prices are flexible, output depends on factors of production and technology, and unemployment remains at its natural rate.

In the **short run**, prices are fixed, output depends on aggregate demand, and unemployment is inversely related to output.

The Keynesian Cross

The Keynesian Cross is a simple closed-economy model developed by **J.M. Keynes**.

It explains that national income is determined by planned expenditure.

Equilibrium occurs when actual expenditure equals planned expenditure.

Important Notations

I = Planned Investment

E = Planned Expenditure

Y = Real GDP (Actual Expenditure)

Actual expenditure equals the value of GDP.

Planned expenditure is the amount households, firms, and the government intend to spend.

Elements of the Keynesian Cross

Consumption depends on disposable income.

Consumption Function:

$$C = C(Y - T)$$

Government spending and taxes are policy variables.

G = Government Spending

T = Taxes

Investment is assumed to be exogenous.

$$I = \bar{I}$$

Planned expenditure is:

$$E = C(Y - T) + I + G$$

Equilibrium condition:

$$Y = E$$

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Planned Expenditure

Planned expenditure consists of:

Consumption.

Investment.

Government spending.

As income increases, planned expenditure also increases because consumption rises.

Equilibrium Income

Equilibrium income is achieved where:

Actual Expenditure = Planned Expenditure

Or

$$Y = E$$

At equilibrium, there is neither excess production nor inventory shortages.

Increase in Government Purchases

An increase in government spending shifts the planned expenditure curve upward.

Initially, firms experience an unplanned fall in inventories.

Firms increase production.

Income rises until a new equilibrium is reached.

Government Purchases Multiplier

The multiplier measures the increase in income caused by an increase in government spending.

Formula:

$$\text{Government Spending Multiplier} = 1 / (1 - \text{MPC})$$

Where:

MPC = Marginal Propensity to Consume

Example:

If **MPC = 0.8**

Multiplier:

$$1 / (1 - 0.8) = 5$$

A Rs.1 increase in government spending increases income by Rs.5.

Why the Multiplier is Greater Than One

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Government spending directly increases income.
Higher income increases consumption.
Higher consumption further increases income.
This repeated process creates the multiplier effect.

Increase in Taxes

Higher taxes reduce disposable income.
Lower disposable income decreases consumption.
Lower consumption reduces planned expenditure.
Firms reduce production.
National income falls until a new equilibrium is reached.

Tax Multiplier

The tax multiplier measures the change in income caused by a change in taxes.

Formula:

$$\text{Tax Multiplier} = -\text{MPC} / (1 - \text{MPC})$$

Example:

If **MPC = 0.8**

Tax Multiplier:

$$-0.8 / 0.2 = -4$$

A Rs.1 increase in taxes decreases income by Rs.4.

Properties of the Tax Multiplier

The tax multiplier is always negative.
Tax increases reduce consumption and national income.
Its absolute value is greater than one.
It is smaller than the government spending multiplier because part of a tax cut is saved rather than spent.

IS Curve

The IS Curve shows all combinations of income (**Y**) and interest rate (**r**) where the goods market is in equilibrium.
Goods market equilibrium occurs when actual expenditure equals planned expenditure.

IS Equation:

$$Y = C(Y - T) + I(r) + G$$

Deriving the IS Curve

A decrease in the interest rate increases investment.
Higher investment raises planned expenditure.

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Higher planned expenditure increases national income.
Therefore, the IS Curve slopes downward.

Important Exam Points

Keynesian theory explains short-run income determination.
Long run assumes flexible prices.
Short run assumes sticky prices.
The Keynesian Cross was developed by **J.M. Keynes**.
Planned expenditure formula: $E = C(Y - T) + I + G$.
Equilibrium condition: $Y = E$.
Consumption depends on disposable income.
Government spending increases equilibrium income.
Government Spending Multiplier: $1 / (1 - MPC)$.
If $MPC = 0.8$, multiplier = 5.
Tax Multiplier: $-MPC / (1 - MPC)$.
If $MPC = 0.8$, tax multiplier = -4.
Government spending multiplier is larger than the tax multiplier.
The IS Curve represents equilibrium in the goods market.
IS equation: $Y = C(Y - T) + I(r) + G$.
Lower interest rates increase investment and income.
The IS Curve slopes downward.

Important MCQs

1. The Keynesian Cross explains that income is determined by:

- A) Technology
- B) Labor supply
- C) Planned expenditure ✓**
- D) Money supply

2. The equilibrium condition in the Keynesian Cross is:

- A) $C = I$
- B) $Y = C$
- C) $Y = E$ ✓**
- D) $G = T$

3. The Government Spending Multiplier is:

- A) MPC
- B) $1 - MPC$
- C) $1 / (1 - MPC)$ ✓**
- D) $MPC / (1 - MPC)$

4. If $MPC = 0.8$, the tax multiplier equals:

- A) 5
- B) 4
- C) -4 ✓**
- D) -5

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5. The IS Curve shows combinations of:

- A) Inflation and unemployment
- B) Output and price level
- C) Interest rate and income ✓
- D) Saving and investment

Short Conceptual Questions

Q1. Why is the Government Spending Multiplier greater than one?

Answer:

An increase in government spending raises income, which increases consumption and creates repeated rounds of spending.

This chain reaction makes the total increase in income larger than the initial spending.

Q2. Why is the Tax Multiplier negative?

Answer:

Higher taxes reduce disposable income and consumer spending.
Lower consumption decreases aggregate demand and national income.

Numerical / Formula Questions

Question 1

If **MPC = 0.75** and government spending increases by **Rs. 200 million**, calculate the increase in national income.

Solution

Government Spending Multiplier:

$$= 1 / (1 - MPC)$$

$$= 1 / (1 - 0.75)$$

$$= 1 / 0.25$$

$$= 4$$

Increase in Income:

$$\Delta Y = \text{Multiplier} \times \Delta G$$

$$= 4 \times 200$$

$$= \text{Rs. 800 million}$$

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Answer: National income increases by Rs. 800 million.

Question 2

If **MPC = 0.80** and taxes increase by **Rs. 100 million**, calculate the change in national income.

Solution

Tax Multiplier:

$$= -\text{MPC} / (1 - \text{MPC})$$

$$= -0.80 / 0.20$$

$$= -4$$

Change in Income:

$$\Delta Y = \text{Tax Multiplier} \times \Delta T$$

$$= -4 \times 100$$

$$= -\text{Rs. 400 million}$$

Answer: National income decreases by Rs. 400 million.

Lesson 27: IS-LM Framework (Summarized Notes)

The IS Curve

The IS curve represents equilibrium in the goods market.

It shows all combinations of income (**Y**) and interest rate (**r**) where planned expenditure equals actual expenditure.

The IS curve is negatively sloped.

A lower interest rate increases investment, raises planned expenditure, and increases output.

IS Curve and Loanable Funds Model

A fall in the interest rate increases investment demand.

Higher investment raises national income.

Thus, the goods market reaches a new equilibrium at a higher level of output.

The IS curve summarizes this inverse relationship between interest rate and income.

Fiscal Policy and the IS Curve

Fiscal policy affects aggregate demand through government spending (**G**) and taxes (**T**).

An increase in government spending shifts the IS curve to the right.

A decrease in taxes also shifts the IS curve to the right.

Expansionary fiscal policy increases output and income.

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Shift in IS Curve due to Government Spending

When government spending increases:
Planned expenditure increases.
Output and income increase.
The entire IS curve shifts to the right.
Higher government spending raises aggregate demand.

Theory of Liquidity Preference

John Maynard Keynes explained that the interest rate is determined by the demand and supply of money.
The money market reaches equilibrium when money demand equals money supply.

Money market equilibrium:

$$M/P = L(r)$$

Where:

M/P = Real Money Supply

$L(r)$ = Money Demand

Money Supply

The central bank controls the nominal money supply.
With a fixed price level, the supply of real money balances remains fixed.
The money supply curve is vertical.

Money Demand

People demand money for transactions and other purposes.
Money demand decreases as the interest rate increases.
Higher interest rates encourage holding interest-bearing assets instead of money.

Money Market Equilibrium

The interest rate adjusts until money demand equals money supply.

Formula:

$$M/P = L(r)$$

At equilibrium, there is neither excess supply nor excess demand for money.

Central Bank and Interest Rate

To increase the interest rate, the central bank reduces the money supply.
A lower money supply decreases real money balances.
The interest rate rises until money market equilibrium is restored.

LM Curve

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The LM curve represents equilibrium in the money market.

It shows combinations of income and interest rate where money demand equals money supply.

Equation:

$$M/P = L(r, Y)$$

Slope of the LM Curve

The LM curve is positively sloped.

Higher income increases the demand for money.

With a fixed money supply, the interest rate must rise to restore equilibrium.

Shift in LM Curve

An increase in the money supply shifts the LM curve to the right (or downward).

A decrease in the money supply shifts the LM curve to the left (or upward).

Short-Run Equilibrium

The economy reaches short-run equilibrium where the IS and LM curves intersect.

At this point, both the goods market and money market are simultaneously in equilibrium.

Goods market:

$$Y = C(Y - T) + I(r) + G$$

Money market:

$$M/P = L(r, Y)$$

Policy Analysis Using the IS-LM Model

The IS-LM model explains how fiscal and monetary policies affect income and interest rates.

Fiscal policy shifts the IS curve.

Monetary policy shifts the LM curve.

Increase in Government Purchases

An increase in government spending shifts the IS curve right.

Output increases.

Higher income increases money demand.

Interest rates rise.

Higher interest rates reduce investment.

This reduction in private investment is called the **crowding-out effect**.

Tax Cut

A tax cut increases disposable income.

Consumption increases.

The IS curve shifts to the right.

Because households save part of the tax cut, its effect is smaller than an equal increase in government spending.

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Expansionary Monetary Policy

An increase in the money supply shifts the LM curve right.
Interest rates fall.
Lower interest rates increase investment.
Investment raises output and income.

Interaction Between Fiscal and Monetary Policy

Fiscal and monetary policies may work together.
The central bank may respond to changes in government spending by changing the money supply.
Different monetary responses produce different economic outcomes.

Central Bank Responses to Higher Government Spending

Response 1: Hold Money Supply Constant

The LM curve does not shift.
Output increases.
Interest rates increase.

Response 2: Hold Interest Rate Constant

The central bank increases the money supply.
The LM curve shifts right.
Output increases more.
Interest rates remain unchanged.

Response 3: Hold Output Constant

The central bank reduces the money supply.
The LM curve shifts left.
Output remains unchanged.
Interest rates increase significantly.

IS and LM Shocks

IS Shocks

IS shocks result from unexpected changes in aggregate demand.

Examples:
Changes in consumer confidence.
Changes in business investment.
Stock market booms or crashes.

LM Shocks

LM shocks result from unexpected changes in money demand.

Examples:
Credit card fraud increases cash demand.
ATMs and digital banking reduce money demand.

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Central Bank Policy Instrument

The central bank usually targets the interest rate rather than the money supply.

It changes the money supply to achieve the desired interest rate.

Interest rates are easier to observe and help stabilize the economy when LM shocks occur frequently.

Important Exam Points

The IS curve represents equilibrium in the goods market.

The LM curve represents equilibrium in the money market.

IS equation: $Y = C(Y - T) + I(r) + G$.

LM equation: $M/P = L(r, Y)$.

The IS curve is negatively sloped.

The LM curve is positively sloped.

Government spending shifts the IS curve to the right.

Tax cuts also shift the IS curve right.

Money supply increases shift the LM curve right.

Money supply decreases shift the LM curve left.

Short-run equilibrium occurs at the intersection of IS and LM.

Expansionary fiscal policy raises output and interest rates.

Expansionary monetary policy lowers interest rates and raises output.

Higher interest rates reduce investment (crowding-out effect).

The central bank controls the money supply.

Money market equilibrium: $M/P = L(r)$.

The central bank often targets interest rates instead of money supply.

IS shocks affect aggregate demand.

LM shocks affect money demand.

Important MCQs

1. The IS curve represents equilibrium in the:

- A) Labor market
- B) Foreign exchange market
- C) **Goods market ✓**
- D) Money market

2. The LM curve represents equilibrium in the:

- A) Goods market
- B) Labor market
- C) **Money market ✓**
- D) Capital market

3. Which policy shifts the LM curve to the right?

- A) Increase in taxes
- B) Decrease in government spending
- C) **Increase in money supply ✓**
- D) Increase in interest rate

4. An increase in government spending causes the IS curve to:

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- A) Shift left
- B) Remain unchanged
- C) Shift right ✓**
- D) Become vertical

5. The crowding-out effect occurs because:

- A) Prices increase
- B) Taxes increase
- C) Higher interest rates reduce private investment ✓**
- D) Money supply decreases automatically

Short Conceptual Questions

Q1. Why is the IS curve downward sloping?

Answer:

Lower interest rates increase investment spending, which raises aggregate demand and output. Therefore, income rises as the interest rate falls.

Q2. Why is the LM curve upward sloping?

Answer:

Higher income increases money demand while money supply remains fixed. The interest rate must rise to restore equilibrium in the money market.

Numerical / Formula Questions

Question 1

If the real money supply is $M/P = 600$ and money demand is $L(r, Y) = 600$, determine whether the money market is in equilibrium.

Solution

Money Market Equilibrium Condition:

$$M/P = L(r, Y)$$

Given:

$$600 = 600$$

Therefore, money demand equals money supply.

Answer: The money market is in equilibrium.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Question 2

If $MPC = 0.75$ and government spending increases by **Rs. 120 million**, calculate the total increase in income using the Government Spending Multiplier.

Solution

Government Spending Multiplier:

$$= 1 / (1 - MPC)$$

$$= 1 / (1 - 0.75)$$

$$= 1 / 0.25$$

$$= 4$$

Increase in Income:

$$\Delta Y = \text{Multiplier} \times \Delta G$$

$$= 4 \times 120$$

$$= \text{Rs. 480 million}$$

Answer: National income increases by Rs. 480 million.

Lesson 29: IS-LM Framework and Aggregate Demand (Summarized Notes)

IS-LM Framework and Aggregate Demand

The IS-LM model explains short-run equilibrium when the price level is fixed.

When the price level changes, the LM curve shifts, changing income and interest rates.

The Aggregate Demand (AD) curve is derived from this relationship between the price level and output.

Deriving the Aggregate Demand (AD) Curve

A higher price level reduces real money balances.

Lower real money balances shift the LM curve to the left.

Interest rates increase.

Higher interest rates reduce investment.

Lower investment decreases output.

Therefore, the AD curve slopes downward.

Relationship:

$$\uparrow P \rightarrow \downarrow (M/P) \rightarrow \text{LM shifts left} \rightarrow \uparrow r \rightarrow \downarrow I \rightarrow \downarrow Y$$

Monetary Policy and the AD Curve

Expansionary monetary policy increases the money supply.

Higher money supply shifts the LM curve to the right.

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Interest rates decrease.
Investment increases.
Output increases at every price level.
As a result, the Aggregate Demand curve shifts to the right.

Sequence:

$\uparrow M \rightarrow \text{LM shifts right} \rightarrow \downarrow r \rightarrow \uparrow I \rightarrow \uparrow Y$

Fiscal Policy and the AD Curve

Expansionary fiscal policy includes higher government spending or lower taxes.
Lower taxes increase consumption.
Higher government spending increases aggregate demand.
The IS curve shifts to the right.
Output rises at every price level.
The AD curve shifts to the right.

Sequence:

$\uparrow G \text{ or } \downarrow T \rightarrow \uparrow C \rightarrow \text{IS shifts right} \rightarrow \uparrow Y$

IS-LM and AD-AS in the Short Run and Long Run

In the short run, prices are sticky.
Changes in aggregate demand mainly affect output.
Over time, prices adjust.
Price adjustment moves the economy toward long-run equilibrium.

If:

$Y > Y^* \rightarrow \text{Prices Rise}$

$Y < Y^* \rightarrow \text{Prices Fall}$

$Y = Y^* \rightarrow \text{Prices Remain Constant}$

Short-Run Effects of a Negative IS Shock

A negative IS shock shifts the IS curve left.
Aggregate demand also shifts left.
Output falls below full-employment output.
Interest rates decline.
The economy enters a recessionary gap.

Long-Run Adjustment after an IS Shock

As prices gradually fall:
Real money balances increase.
The LM curve shifts right.
Interest rates fall further.
Investment rises.

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Output gradually returns to full-employment level.
Long-run equilibrium is restored.

Short-Run Effects of an Expansionary Fiscal Policy

Government spending increases or taxes decrease.
Output increases.
Prices remain unchanged because they are sticky.
Interest rates increase.
Consumption increases because income rises.
Investment decreases due to higher interest rates (crowding-out effect).

Long-Run Effects of Expansionary Fiscal Policy

Prices rise gradually.
The LM curve shifts left due to lower real money balances.
Output returns to its natural level.
Interest rates remain higher than before.
Consumption returns to its original level.
Investment decreases further because of higher interest rates.

Expansionary Monetary Policy

The central bank increases the money supply.
The LM curve shifts right.
Interest rates fall.
Investment increases.
Output increases in the short run.
Aggregate demand shifts right.

Short-Run Effects of Expansionary Monetary Policy

Output increases.
Prices remain fixed.
Interest rates decrease.
Consumption increases.
Investment increases because borrowing becomes cheaper.

Long-Run Effects of Expansionary Monetary Policy

As prices rise:
Real money balances decrease.
The LM curve shifts back to the left.
Output returns to full-employment.
Interest rates return to their original level.
Consumption returns to its original level.
Investment also returns to its original level.
Only the price level remains permanently higher.

Key Conclusion

Expansionary monetary policy affects output only in the short run.
In the long run, money is neutral.
The permanent effect of increasing the money supply is a higher price level.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Important Exam Points

The AD curve is derived from the IS-LM model.
The AD curve slopes downward.
Higher prices reduce real money balances.
Lower real money balances shift the LM curve left.
Higher interest rates reduce investment.
Expansionary monetary policy shifts the LM curve right.
Expansionary fiscal policy shifts the IS curve right.
Higher government spending increases aggregate demand.
Lower taxes increase consumption.
Negative IS shocks reduce output in the short run.
Price adjustment restores long-run equilibrium.
Short-run prices are sticky.
Long-run prices are flexible.
Expansionary fiscal policy increases output and interest rates in the short run.
Expansionary monetary policy lowers interest rates and raises output in the short run.
In the long run, output returns to full-employment.
Money is neutral in the long run.
The only permanent effect of higher money supply is a higher price level.

Important MCQs

1. The Aggregate Demand curve is derived from the relationship between:

- A) Inflation and unemployment
- B) Saving and investment
- C) Price level and output ✓**
- D) Consumption and income

2. A rise in the price level causes the LM curve to:

- A) Shift right
- B) Shift left ✓**
- C) Become horizontal
- D) Remain unchanged

3. Expansionary monetary policy shifts the:

- A) IS curve left
- B) AD curve left
- C) LM curve right ✓**
- D) LRAS curve right

4. In the long run, an increase in the money supply permanently increases:

- A) Output
- B) Consumption
- C) Investment
- D) Price level ✓**

5. Expansionary fiscal policy shifts the:

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- A) LM curve left
 - B) SRAS curve left
 - C) IS curve right ✓**
 - D) LRAS curve right
-

Short Conceptual Questions

Q1. Why does the Aggregate Demand curve slope downward?

Answer:

A higher price level reduces real money balances, increasing interest rates and reducing investment. Lower investment decreases aggregate output, creating a downward-sloping AD curve.

Q2. Why does monetary policy affect output only in the short run?

Answer:

Because prices are sticky in the short run, output changes with aggregate demand. In the long run, prices adjust and output returns to its natural level.

Numerical / Formula Questions

Question 1

Suppose the real money supply decreases from **500** to **400** due to a rise in the price level. What happens to the LM curve and interest rate?

Solution

Real money balances:

M/P decreases

Therefore:

LM curve shifts left

As the LM curve shifts left:

Interest rate increases

Answer: The LM curve shifts left and the interest rate rises.

Question 2

If government spending increases by **Rs. 150 million** and the Government Spending Multiplier is **4**, calculate the increase in national income.

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Solution

Formula:

$$\Delta Y = \text{Multiplier} \times \Delta G$$

$$\Delta Y = 4 \times 150$$

$$\Delta Y = \text{Rs. 600 million}$$

Answer: National income increases by **Rs. 600 million**.

Lesson 30: The Mundell–Fleming Model (Summarized Notes)

The Mundell–Fleming Model

The Mundell–Fleming Model is an extension of the IS-LM model for a **small open economy**. It explains the relationship between the **nominal exchange rate** and **national output**. The model assumes **perfect capital mobility**, meaning capital moves freely across countries. The domestic interest rate is equal to the world interest rate.

$$r = r^* \text{ (World Interest Rate)}$$

Key Assumptions

The economy is small compared to the world economy.
Capital is perfectly mobile.
The domestic interest rate equals the world interest rate.
International trade affects national income.
Exchange rates play an important role in determining output.

IS* Curve (Goods Market Equilibrium)

The IS* curve represents equilibrium in the goods market of an open economy.

Equation:

$$Y = C(Y - T) + I(r^*) + G + NX(e)$$

Where:

Y = National Income

C = Consumption

I = Investment

G = Government Spending

NX = Net Exports

e = Nominal Exchange Rate

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A depreciation of the domestic currency increases net exports.
Higher net exports increase aggregate demand and national income.
Therefore, the IS* curve is downward sloping.

Relationship:

$$\downarrow e \rightarrow \uparrow NX \rightarrow \uparrow Y$$

LM* Curve (Money Market Equilibrium)

The LM* curve represents equilibrium in the money market.

Equation:

$$M/P = L(r^*, Y)$$

Since the interest rate is fixed at the world level, only one level of income satisfies money market equilibrium.
Therefore, the LM* curve is vertical.

Equilibrium in the Mundell–Fleming Model

The economy reaches equilibrium where the IS* and LM* curves intersect.
This determines:
The equilibrium exchange rate.
The equilibrium level of national income.

Exchange Rate Systems

Floating Exchange Rate

The exchange rate is determined by market demand and supply.
The central bank does not fix the exchange rate.
Exchange rates fluctuate freely according to economic conditions.

Fixed Exchange Rate

The central bank maintains a predetermined exchange rate.
It buys or sells foreign currency to keep the exchange rate constant.

Fiscal Policy under Floating Exchange Rates

Expansionary fiscal policy increases government spending or decreases taxes.
The IS* curve shifts to the right.
The exchange rate appreciates.
Net exports decrease.
The increase in net exports is completely crowded out.
National income remains unchanged.

Results:

$$\Delta e > 0$$

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$$\Delta Y = 0$$

Lessons about Fiscal Policy

In a small open economy with perfect capital mobility:
Fiscal policy cannot increase real GDP under floating exchange rates.
Government spending causes exchange rate appreciation.
Higher exchange rates reduce net exports.
Fiscal expansion crowds out net exports rather than investment.

Crowding-Out Effect

Closed Economy

Higher government spending raises interest rates.
Higher interest rates reduce investment.

Small Open Economy

Higher government spending appreciates the exchange rate.
Higher exchange rates reduce net exports.
Investment remains unchanged because the interest rate is fixed at the world level.

Monetary Policy under Floating Exchange Rates

Expansionary monetary policy increases the money supply.
The LM* curve shifts to the right.
The domestic currency depreciates.
Net exports increase.
Output and income rise.

Results:

$$\Delta e < 0$$

$$\Delta Y > 0$$

Lessons about Monetary Policy

Monetary policy is highly effective under floating exchange rates.
Increasing the money supply depreciates the domestic currency.
Currency depreciation makes exports cheaper.
Net exports increase.
Higher net exports increase aggregate demand and national income.

Comparison with Closed Economy

Closed Economy:

$$\uparrow M \rightarrow \downarrow r \rightarrow \uparrow I \rightarrow \uparrow Y$$

Monetary policy increases investment through lower interest rates.

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Small Open Economy:

$$\uparrow M \rightarrow \downarrow e \rightarrow \uparrow NX \rightarrow \uparrow Y$$

Monetary policy increases output through higher net exports.

Key Conclusion

Under floating exchange rates:

Fiscal policy is ineffective in changing output.

Monetary policy is effective because it influences the exchange rate and net exports.

Expansionary monetary policy shifts demand from foreign goods to domestic goods rather than increasing world demand.

Important Exam Points

The Mundell–Fleming Model is an extension of the IS-LM model.

It applies to a small open economy.

The model assumes perfect capital mobility.

Domestic interest rate equals world interest rate ($r = r^*$).

IS* represents goods market equilibrium.

LM* represents money market equilibrium.

IS* equation: $Y = C(Y - T) + I(r^*) + G + NX(e)$.

LM* equation: $M/P = L(r^*, Y)$.

The IS* curve is downward sloping.

The LM* curve is vertical.

Floating exchange rates are market determined.

Fixed exchange rates are controlled by the central bank.

Fiscal policy appreciates the exchange rate under floating rates.

Fiscal policy does not change output under floating exchange rates.

Expansionary monetary policy depreciates the currency.

Currency depreciation increases net exports.

Monetary policy is effective under floating exchange rates.

Closed economy monetary policy works through investment.

Open economy monetary policy works through net exports.

Important MCQs

1. The Mundell–Fleming Model is an extension of the:

- A) Solow Model
- B) AD-AS Model
- C) IS-LM Model ✓**
- D) Keynesian Cross

2. The Mundell–Fleming Model assumes:

- A) Closed economy
- B) Fixed interest rate by government
- C) Perfect capital mobility ✓**
- D) Zero inflation

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

3. Under floating exchange rates, expansionary fiscal policy causes:

- A) Currency depreciation
- B) Higher output
- C) **Currency appreciation ✓**
- D) Lower government spending

4. Under floating exchange rates, expansionary monetary policy causes:

- A) Exchange rate appreciation
- B) Lower net exports
- C) **Currency depreciation and higher output ✓**
- D) Lower consumption

5. In the Mundell–Fleming Model, the LM* curve is:

- A) Downward sloping
- B) Horizontal
- C) **Vertical ✓**
- D) Backward sloping

Short Conceptual Questions

Q1. Why is fiscal policy ineffective under floating exchange rates?

Answer:

Expansionary fiscal policy appreciates the domestic currency, reducing net exports. This completely offsets the increase in government spending, leaving output unchanged.

Q2. Why is monetary policy effective under floating exchange rates?

Answer:

An increase in the money supply depreciates the domestic currency. The depreciation increases net exports, raising aggregate demand and national income.

Numerical / Formula Questions

Question 1

Suppose expansionary monetary policy causes net exports to increase by **Rs. 250 million**. If the increase in net exports directly raises national income by the same amount, calculate the increase in GDP.

Solution

Given:

Increase in Net Exports (ΔNX) = Rs. 250 million

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Since:

$$\Delta Y = \Delta NX$$

$$\Delta Y = \text{Rs. 250 million}$$

Answer: National income (GDP) increases by **Rs. 250 million**.

Question 2

A country's exchange rate falls from **120 units** of foreign currency per domestic currency to **110 units** after expansionary monetary policy. State the effect on net exports and output according to the Mundell–Fleming Model.

Solution

A lower exchange rate means **currency depreciation**.

Therefore:

$$\downarrow e \rightarrow \uparrow NX \rightarrow \uparrow Y$$

Answer: Net exports increase and national output rises.

Lesson 31: The Mundell–Fleming Model (Continued) (Summarized Notes)

Trade Policy under Floating Exchange Rates

Trade policy includes import tariffs and import quotas.

A tariff or quota reduces imports and increases net exports.

The IS* curve shifts to the right.

However, the exchange rate appreciates, reducing exports.

As a result, national income remains unchanged.

Lessons about Trade Policy

Import restrictions cannot permanently reduce a trade deficit.

Imports decrease, but exports also decrease due to exchange rate appreciation.

Total net exports remain unchanged.

Trade restrictions reduce the gains from international trade.

They protect some domestic industries but harm export industries.

Import restrictions create sectoral shifts, increasing frictional unemployment.

Fixed Exchange Rate System

Under a fixed exchange rate system, the central bank maintains a predetermined exchange rate.

The central bank buys or sells domestic currency whenever necessary.

It adjusts the money supply to keep the exchange rate fixed.

The nominal exchange rate remains constant, although the real exchange rate may change in the long run.

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Equilibrium under Fixed Exchange Rates

If the market exchange rate is above the fixed rate, the central bank intervenes by selling domestic currency.

If the market exchange rate is below the fixed rate, the central bank buys domestic currency.

These actions shift the LM* curve to maintain the fixed exchange rate.

Fiscal Policy under Fixed Exchange Rates

Expansionary fiscal policy increases aggregate demand.

The IS* curve shifts to the right.

The exchange rate tends to appreciate.

To prevent appreciation, the central bank increases the money supply.

The LM* curve shifts right.

Output increases while the exchange rate remains fixed.

Results:

$$\Delta e = 0$$

$$\Delta Y > 0$$

Effectiveness of Fiscal Policy

Fiscal policy is highly effective under fixed exchange rates.

The central bank supports fiscal expansion by increasing the money supply.

Both fiscal policy and monetary expansion increase output together.

Monetary Policy under Fixed Exchange Rates

Expansionary monetary policy initially shifts the LM* curve right.

The exchange rate tends to depreciate.

To maintain the fixed exchange rate, the central bank buys domestic currency.

The money supply returns to its original level.

The LM* curve shifts back to its initial position.

Output remains unchanged.

Results:

$$\Delta e = 0$$

$$\Delta Y = 0$$

Effectiveness of Monetary Policy

Monetary policy is ineffective under fixed exchange rates.

Any attempt to change the money supply is offset by central bank intervention.

The exchange rate remains fixed and output does not change.

Trade Policy under Fixed Exchange Rates

Import restrictions reduce imports.

The exchange rate tends to appreciate.

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The central bank increases the money supply to maintain the fixed exchange rate.
The LM* curve shifts right.
National income and net exports increase.

Results:

$$\Delta e = 0$$

$$\Delta Y > 0$$

$$\Delta NX > 0$$

These gains occur mainly at the expense of foreign countries because demand shifts from imported goods to domestic goods.

Summary of Policy Effects

Floating Exchange Rates

Fiscal policy does not change output.
Monetary policy increases output.
Import restrictions do not affect output or net exports.

Fixed Exchange Rates

Fiscal policy increases output.
Monetary policy has no effect on output.
Import restrictions increase both output and net exports.

Interest Rate Differentials

The domestic interest rate may differ from the world interest rate for two reasons.

1. Country Risk

Country risk is the possibility that borrowers may fail to repay loans due to political or economic instability.
Higher country risk increases the interest rate demanded by lenders.

2. Expected Exchange Rate Changes

If the domestic currency is expected to depreciate, lenders demand higher interest rates to compensate for expected losses.

Risk Premium

The domestic interest rate is:

$$r = r^* + \theta$$

Where:

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r = Domestic Interest Rate

r^* = World Interest Rate

θ = Risk Premium

Higher risk premium raises borrowing costs.

Effects of an Increase in Risk Premium (θ)

A higher risk premium increases domestic interest rates.

Investment decreases.

The IS* curve shifts left.

Money demand decreases.

The LM* curve shifts right.

The domestic currency depreciates.

Net exports increase.

The increase in net exports is larger than the decline in investment.

National income increases.

Key Conclusion

Under floating exchange rates:

Monetary policy is effective.

Fiscal policy is ineffective.

Under fixed exchange rates:

Fiscal policy is effective.

Monetary policy is ineffective.

Country risk and expected currency depreciation increase domestic interest rates.

Higher risk premium causes currency depreciation and higher net exports.

Important Exam Points

The Mundell–Fleming Model analyzes open economies.

Trade policy under floating exchange rates does not change output.

Import restrictions reduce imports but also reduce exports.

Fixed exchange rates are maintained by central bank intervention.

The central bank buys or sells domestic currency to keep exchange rates fixed.

Fiscal policy is ineffective under floating exchange rates.

Fiscal policy is effective under fixed exchange rates.

Monetary policy is effective under floating exchange rates.

Monetary policy is ineffective under fixed exchange rates.

Import restrictions increase output under fixed exchange rates.

Risk premium is represented by θ .

Domestic interest rate equation: $r = r^* + \theta$.

Country risk raises domestic interest rates.

Expected currency depreciation also raises domestic interest rates.

Higher risk premium shifts the IS* curve left.

Higher risk premium shifts the LM* curve right.

Currency depreciation increases net exports.

Higher net exports may increase national income.

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Important MCQs

1. Under floating exchange rates, fiscal policy is:

- A) Highly effective
- B) Partially effective
- C) Ineffective in changing output ✓**
- D) Effective only with higher taxes

2. Under fixed exchange rates, monetary policy is:

- A) Highly effective
- B) Moderately effective
- C) Ineffective ✓**
- D) Effective only in the long run

3. The domestic interest rate with a risk premium is:

- A) $r = r^* - \theta$
- B) $r = \theta - r^*$
- C) $r = r^* + \theta$ ✓**
- D) $r = \theta \times r^*$

4. Country risk causes:

- A) Lower interest rates
- B) Currency appreciation only
- C) Higher interest rates ✓**
- D) Lower money supply automatically

5. Under fixed exchange rates, expansionary fiscal policy results in:

- A) $\Delta Y = 0$
- B) $\Delta e > 0$
- C) $\Delta Y > 0$ and $\Delta e = 0$ ✓**
- D) $\Delta NX = 0$

Short Conceptual Questions

Q1. Why is monetary policy ineffective under fixed exchange rates?

Answer:

The central bank offsets any change in the money supply to maintain the fixed exchange rate. Therefore, output and income remain unchanged.

Q2. What is country risk in the Mundell–Fleming Model?

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Answer:

Country risk is the possibility that borrowers may default due to political or economic instability. Lenders demand higher interest rates as compensation for this risk.

Numerical / Formula Questions

Question 1

Suppose the world interest rate is 5% and the country's risk premium is 3%. Calculate the domestic interest rate.

Solution

Formula:

$$r = r^* + \theta$$

Given:

$$r^* = 5\%$$

$$\theta = 3\%$$

$$r = 5\% + 3\%$$

$$r = 8\%$$

Answer: The domestic interest rate is 8%.

Question 2

A country has a world interest rate of 6% and its risk premium increases from 2% to 4%. Calculate the new domestic interest rate.

Solution

Formula:

$$r = r^* + \theta$$

New Domestic Interest Rate:

$$r = 6\% + 4\%$$

$$r = 10\%$$

Answer: The new domestic interest rate is 10%.

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Lesson 32: The Mundell-Fleming Model (Continued) & The Three Models of Aggregate Supply (Summarized Notes)

Why Income Might Not Rise

Even after currency depreciation, national income may fail to increase because the central bank may reduce the money supply to stop depreciation. Higher import prices may increase the price level and reduce real money balances. Consumers may also demand more money due to uncertainty, shifting the LM* curve left and lowering output.

The South East Asian Crisis

The Asian Financial Crisis caused severe depreciation of exchange rates, stock market crashes, and declines in GDP across countries like Indonesia, Malaysia, South Korea, Thailand, and Japan. The crisis demonstrated the impact of financial instability on economic growth.

Floating vs Fixed Exchange Rates

Floating exchange rates allow the exchange rate to adjust freely according to market forces. This gives the central bank flexibility to use monetary policy for economic stabilization.

Fixed exchange rates keep the currency value constant against another currency. This reduces exchange rate uncertainty, encourages international trade, and controls inflation by limiting excessive money creation.

Mundell-Fleming Model and Aggregate Demand

The Aggregate Demand (AD) curve is derived from the Mundell-Fleming model by allowing the price level to change.

The IS* equation is:

$$Y = C(Y - T) + I(r) + G + NX(\epsilon)^*$$

The LM* equation is:

$$M/P = L(r, Y)^*$$

An increase in the price level reduces real money balances (M/P), shifts the LM* curve left, appreciates the real exchange rate, reduces net exports, and decreases national income. Therefore, the AD curve slopes downward.

From Short Run to Long Run

When output falls below its natural level, prices gradually decline over time. Falling prices increase real money balances, shift LM* right, increase net exports, and restore output to its natural level.

Large Open Economy

A large open economy lies between a closed economy and a small open economy.

Expansionary monetary policy lowers the interest rate, increases investment, depreciates the exchange rate, raises net exports, and increases output. However, these effects are smaller than in a small open economy.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Three Models of Aggregate Supply

Economists explain short-run aggregate supply using three major models:

Sticky-Wage Model

Imperfect-Information Model

Sticky-Price Model

All three models lead to the Short-Run Aggregate Supply equation:

$$Y = \bar{Y} + \alpha(P - P_e)$$

Where:

Y = Actual Output

$\bar{Y}$ = Natural Output

α = Positive Constant

P = Actual Price Level

P_e = Expected Price Level

Sticky-Wage Model

Workers and firms agree on nominal wages before the actual price level is known.

Nominal Wage Formula:

$$W = \omega \times P_e$$

If the actual price equals the expected price, output remains at its natural level.

If actual prices become higher than expected, real wages fall, firms hire more workers, and output rises.

If actual prices become lower than expected, real wages increase, employment falls, and output declines.

Prediction of Sticky-Wage Model

The model predicts that real wages move opposite to business cycles.

During economic booms, real wages should decrease.

During recessions, real wages should increase.

However, empirical evidence shows this prediction is generally not supported in the real world.

Important Exam Points

The Mundell-Fleming model derives the Aggregate Demand curve for an open economy.

A rise in the price level decreases real money balances and shifts the LM^* curve left.

The AD curve is downward sloping because higher prices reduce net exports and output.

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Floating exchange rates provide monetary policy independence.

Fixed exchange rates promote exchange rate stability and lower inflation.

A large open economy combines features of both closed and small open economies.

The three Aggregate Supply models are Sticky-Wage, Imperfect-Information, and Sticky-Price models.

Short-run Aggregate Supply equation:

$$Y = \bar{Y} + \alpha(P - P_e)$$

Sticky-Wage Model assumes wages are fixed before actual prices are known.

If $P > P_e$, output increases above the natural level.

If $P < P_e$, output falls below the natural level.

Important MCQs

1. Which model explains aggregate supply by assuming nominal wages are predetermined?

- A. Solow Growth Model
- B. IS-LM Model
- C. Sticky-Wage Model ✓
- D. Phillips Curve

2. In the Mundell-Fleming model, an increase in the price level causes:

- A. LM^* shifts right
- B. LM shifts left ✓*
- C. IS^* shifts right
- D. Net exports increase

3. The Aggregate Demand curve slopes downward because:

- A. Investment always rises
- B. Government spending falls

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C. Higher prices reduce real money balances and output ✓

D. Consumption always increases

4. Which exchange rate system gives greater freedom to monetary policy?

A. Fixed Exchange Rate

B. Floating Exchange Rate ✓

C. Pegged Exchange Rate

D. Currency Board

5. According to the Sticky-Wage Model, when actual prices exceed expected prices:

A. Employment falls

B. Output decreases

C. Firms hire more workers and output rises ✓

D. Investment declines

Short Conceptual Questions

Q1. Why is the Aggregate Demand curve downward sloping in the Mundell-Fleming model?

Answer:

A higher price level reduces real money balances, shifts the LM* curve left, decreases net exports, and lowers national income.

Q2. Why does the Sticky-Wage Model predict higher output when prices rise unexpectedly?

Answer:

Higher-than-expected prices reduce real wages, encouraging firms to hire more workers and produce more output.

Numerical / Formula Questions

Question 1

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Given:

Natural Output ($\bar{Y}$) = 500

$\alpha = 4$

Actual Price Level (P) = 120

Expected Price Level (P_e) = 115

Find Actual Output (Y).

Solution

Formula:

$$Y = \bar{Y} + \alpha(P - P_e)$$

$$Y = 500 + 4(120 - 115)$$

$$Y = 500 + 20$$

Answer: Y = 520

Question 2

Given:

Expected Price Level (P_e) = 100

Target Real Wage (ω) = 8

Find the Nominal Wage (W).

Solution

Formula:

$$W = \omega \times P_e$$

$$W = 8 \times 100$$

Answer: W = 800

Lesson 33: Three Models of Aggregate Supply (Continued) (Summarized Notes)

The Imperfect-Information Model

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The Imperfect-Information Model assumes that all wages and prices are perfectly flexible and all markets remain in equilibrium. Each producer supplies one good but consumes many goods. Producers know the price of their own product but do not know the overall price level when making production decisions.

Each producer bases production on the **relative price** of their product, which is the product price divided by the overall price level. Since producers do not know the actual price level, they use the expected price level (P_e).

If the actual price level rises above the expected price level ($P > P_e$), producers mistakenly believe that the price of their own product has increased relative to others. As a result, they increase production. When many producers react this way, total output rises.

The Sticky-Price Model

The Sticky-Price Model explains why some firms cannot immediately adjust prices when economic conditions change.

Reasons for Sticky Prices

Long-term contracts prevent frequent price changes.

Menu costs make changing prices expensive.

Firms avoid frequent price changes to maintain customer satisfaction.

Assumptions of the Sticky-Price Model

Some firms can change prices freely, while others must keep prices fixed for a period.

Flexible-price firms set prices according to current economic conditions.

Sticky-price firms determine prices before knowing the actual price level and output.

Desired Price of Flexible Firms

Flexible firms determine prices using the formula:

$$p = P + a(Y - \bar{Y})$$

Where:

p = Desired price of the firm

P = Overall price level

Y = Actual output

$\bar{Y}$ = Natural output

a = Positive constant

Price Set by Sticky-Price Firms

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Sticky-price firms set prices according to expected values.

$$p = P_e + a(Y_e - \bar{Y})$$

Since firms expect output to remain at its natural level,

$$Y_e = \bar{Y}$$

Therefore,

$$p = P_e$$

Overall Price Level

If s is the fraction of firms with sticky prices, the aggregate price level becomes:

$$P = sP_e + (1 - s)[P + a(Y - \bar{Y})]$$

This equation combines prices of sticky-price firms and flexible-price firms.

Aggregate Supply Equation

After simplification, the Short-Run Aggregate Supply equation becomes:

$$Y = \bar{Y} + \alpha(P - P_e)$$

Where:

$$\alpha = ((1 - s)/s) \times a$$

Output rises whenever the actual price level exceeds the expected price level.

Effects of Expected Prices

Higher expected prices (P_e) cause firms with sticky prices to set higher prices in advance.

Flexible-price firms respond by increasing their own prices.

Therefore, higher expected prices lead to a higher overall price level.

Effects of Higher Output

When output rises, demand for goods increases.

Flexible-price firms increase prices.

The larger the proportion of flexible-price firms, the stronger the relationship between output and prices.

Real Wage Prediction

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Unlike the Sticky-Wage Model, the Sticky-Price Model predicts a **pro-cyclical real wage**.

During economic expansion, both output and real wages increase.

During recession, firms reduce production and labor demand, causing real wages to fall.

Important Exam Points

The Imperfect-Information Model assumes perfectly flexible wages and prices.

Producers know only the price of their own product, not the overall price level.

If $P > P_e$, producers mistakenly increase production.

The Sticky-Price Model explains short-run output fluctuations due to slow price adjustments.

Major causes of sticky prices are long-term contracts, menu costs, and customer relationships.

Flexible-price firms adjust prices immediately according to market conditions.

Sticky-price firms set prices based on expected prices.

Aggregate Supply equation:

$$Y = \bar{Y} + \alpha(P - P_e)$$

Sticky-Price Model predicts **pro-cyclical real wages**.

Higher expected prices result in higher actual prices.

Important MCQs

1. In the Imperfect-Information Model, producers know:

- A. Overall price level only
- B. The price of their own product only ✓**
- C. Future inflation rate
- D. Government expenditure

2. According to the Imperfect-Information Model, output increases when:

- A. $P = P_e$

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B. $P < P_e$

C. $P > P_e$ ✓

D. Prices remain constant

3. Which is NOT a reason for sticky prices?

A. Menu costs

B. Long-term contracts

C. Customer satisfaction

D. Perfect competition ✓

4. The Sticky-Price Model assumes that some firms:

A. Never produce goods

B. Have perfectly flexible wages only

C. Cannot immediately change prices ✓

D. Ignore consumer demand

5. The Sticky-Price Model predicts that real wages are:

A. Counter-cyclical

B. Constant

C. Pro-cyclical ✓

D. Independent of output

Short Conceptual Questions

Q1. Why does output increase when actual prices exceed expected prices in the Imperfect-Information Model?

Answer:

Producers believe only their own product price has increased, so they increase production even though the rise is economy-wide.

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Q2. Why do firms keep prices fixed in the Sticky-Price Model?

Answer:

Long-term contracts, menu costs, and customer satisfaction make firms reluctant to change prices frequently.

Numerical / Formula Questions

Question 1

Given:

Natural Output ($\bar{Y}$) = 600

$\alpha = 3$

Actual Price Level (P) = 110

Expected Price Level (P_e) = 105

Find Actual Output (Y).

Solution

Formula:

$$Y = \bar{Y} + \alpha(P - P_e)$$

$$Y = 600 + 3(110 - 105)$$

$$Y = 600 + 15$$

Answer: $Y = 615$

Question 2

Given:

$s = 0.40$

$a = 2$

Find α .

Solution

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Formula:

$$\alpha = ((1 - s)/s) \times a$$

$$\alpha = ((1 - 0.40)/0.40) \times 2$$

$$\alpha = (0.60/0.40) \times 2$$

$$\alpha = 1.5 \times 2$$

Answer: $\alpha = 3$

Lesson 34: Inflation, Unemployment, and the Phillips Curve (Summarized Notes)

Short-Run Aggregate Supply and Inflation

All three Aggregate Supply models imply that output depends on the difference between the actual and expected price levels.

SRAS Equation:

$$Y = \bar{Y} + \alpha(P - P_e)$$

When $P > P_e$, output rises above its natural level.

When $P = P_e$, output equals its natural level.

When $P < P_e$, output falls below its natural level.

A positive Aggregate Demand shock raises output and prices in the short run. As people adjust their expectations, expected prices increase, the SRAS curve shifts upward, and output returns to its natural level.

The Phillips Curve

The Phillips Curve explains the relationship between inflation and unemployment.

Modern Phillips Curve Equation:

$$\pi = \pi_e - \beta(u - u_n) + v$$

Where:

π = Actual inflation rate

π_e = Expected inflation rate

u = Actual unemployment rate

u_n = Natural unemployment rate

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

β = Positive constant

v = Supply shock

Inflation depends on expected inflation, cyclical unemployment, and supply shocks.

Derivation of the Phillips Curve

The Phillips Curve is derived from the Short-Run Aggregate Supply equation.

First, a supply shock is added:

$$P = P^e + (1/\alpha)(Y - \bar{Y}) + v$$

Converting price levels into inflation gives:

$$\pi = \pi^e + (1/\alpha)(Y - \bar{Y}) + v$$

Using **Okun's Law**:

$$(1/\alpha)(Y - \bar{Y}) = -\beta(u - u_n)$$

Substituting this relationship gives the Phillips Curve:

$$\pi = \pi^e - \beta(u - u_n) + v$$

Relationship Between SRAS and Phillips Curve

The SRAS curve shows how output responds to unexpected price changes.

The Phillips Curve shows how unemployment responds to unexpected inflation.

Both describe the short-run effects of demand and supply shocks.

Adaptive Expectations

Adaptive Expectations assume people form future inflation expectations using recently observed inflation.

Formula:

$$\pi^e = \pi_{-1}$$

Substituting into the Phillips Curve:

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$$\pi = \pi_{-1} - \beta(u - u_n) + v$$

Current inflation depends heavily on last year's inflation.

Inflation Inertia

Inflation has inertia because expected inflation is based on past inflation.

Without supply shocks or cyclical unemployment, inflation tends to continue at its current rate.

Past inflation influences wage contracts and price-setting behavior.

Types of Inflation

Cost-Push Inflation

Cost-push inflation results from adverse supply shocks that increase production costs.

Examples include rising oil prices, wage increases, and higher production costs.

Demand-Pull Inflation

Demand-pull inflation occurs when aggregate demand increases rapidly.

Higher demand lowers unemployment below its natural rate and raises inflation.

Short-Run Phillips Curve

In the short run, there is a trade-off between inflation and unemployment.

Lower unemployment usually comes with higher inflation.

Higher unemployment is associated with lower inflation.

Shifting the Phillips Curve

The Phillips Curve shifts whenever expected inflation changes.

An increase in expected inflation shifts the Short-Run Phillips Curve upward.

The trade-off between inflation and unemployment exists only in the short run.

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Sacrifice Ratio

The Sacrifice Ratio measures the percentage of one year's GDP that must be lost to reduce inflation by one percentage point.

Typical value:

Sacrifice Ratio = 5

If inflation falls from **6% to 2%**, inflation decreases by **4 percentage points**.

GDP loss:

$4 \times 5 = 20\%$ of one year's GDP

Rational Expectations

Rational Expectations assume people use all available information, including government policies, to predict future inflation.

People adjust expectations immediately if policy announcements are credible.

Painless Disinflation

If the central bank credibly announces lower inflation, expected inflation falls immediately.

Inflation can decline without increasing unemployment.

Therefore, the sacrifice ratio may become very small.

Natural Rate Hypothesis

Changes in aggregate demand affect output and employment only in the short run.

In the long run, output, employment, and unemployment return to their natural levels.

Hysteresis

Hysteresis means temporary recessions can permanently increase the natural unemployment rate.

Long-term unemployment reduces workers' skills and bargaining power.

Some unemployed workers become structurally unemployed even after the economy recovers.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Important Exam Points

The SRAS equation is $Y = \bar{Y} + \alpha(P - P^e)$.

The modern Phillips Curve is $\pi = \pi^e - \beta(u - u_n) + v$.

Inflation depends on expected inflation, unemployment gap, and supply shocks.

Okun's Law connects output fluctuations with unemployment.

Adaptive Expectations assume future inflation equals past inflation.

Inflation inertia means inflation tends to persist over time.

Cost-push inflation results from supply shocks.

Demand-pull inflation results from higher aggregate demand.

The short-run Phillips Curve shows a trade-off between inflation and unemployment.

The Phillips Curve shifts when expected inflation changes.

Sacrifice Ratio measures GDP lost to reduce inflation.

Rational Expectations use all available information to predict inflation.

Natural Rate Hypothesis states demand affects output only in the short run.

Hysteresis suggests recessions may permanently raise unemployment.

Important MCQs

1. The modern Phillips Curve states that inflation depends on:

- A. Only money supply
 - B. Only unemployment
 - C. Expected inflation, unemployment gap, and supply shocks ✓**
 - D. Government spending only
-

2. According to Adaptive Expectations:

- A. Future inflation is always zero.

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B. Expected inflation equals last year's inflation. ✓

C. Inflation depends only on GDP.

D. Inflation is random.

3. Cost-push inflation is mainly caused by:

A. Increase in aggregate demand

B. Tax reduction

C. Adverse supply shocks ✓

D. Lower wages

4. The Sacrifice Ratio measures:

A. Tax revenue loss

B. Government expenditure

C. GDP loss required to reduce inflation by 1% ✓

D. Increase in unemployment rate

5. Hysteresis means:

A. Inflation permanently increases.

B. GDP always grows.

C. Temporary recessions may permanently increase unemployment. ✓

D. Prices never change.

Short Conceptual Questions

Q1. What is inflation inertia?

Answer:

Inflation inertia means current inflation tends to continue because people base future expectations on past inflation.

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Q2. Why does the Phillips Curve shift upward?

Answer:

An increase in expected inflation raises current inflation at every unemployment level, shifting the curve upward.

Numerical / Formula Questions

Question 1

Given:

Expected Inflation (π_e) = 5%

$\beta = 0.5$

Actual Unemployment (u) = 4%

Natural Unemployment (u_n) = 6%

Supply Shock (v) = 1%

Find the inflation rate (π).

Solution

Formula:

$$\pi = \pi_e - \beta(u - u_n) + v$$

$$\pi = 5 - 0.5(4 - 6) + 1$$

$$\pi = 5 - 0.5(-2) + 1$$

$$\pi = 5 + 1 + 1$$

Answer: $\pi = 7\%$

Question 2

Suppose inflation decreases from 8% to 3%.

Sacrifice Ratio = 5

Find the percentage loss in GDP.

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Solution

Reduction in inflation = $8 - 3 = 5\%$

GDP Loss = **Sacrifice Ratio** $\times$ **Reduction in Inflation**

GDP Loss = $5 \times 5 = 25\%$

Answer: GDP Loss = 25% of one year's GDP

Lesson 35: Government Debt (Summarized Notes)

Government Debt and the Annual Budget Deficit

When a government spends more than it collects in taxes, it borrows from the private sector to finance the budget deficit.

Government debt is the accumulation of all past annual budget deficits.

Budget deficits increase government borrowing, while budget surpluses reduce debt.

Components of Domestic Debt

Domestic debt is divided into three major categories.

Permanent Debt

Permanent debt includes long-term borrowing instruments such as:

Market Loans

Federal Government Bonds

Income Tax Bonds

National Fund Bonds

Federal Investment Bonds

Prize Bonds

Floating Debt

Floating debt includes short-term borrowing instruments such as:

Treasury Bills

Market Treasury Bills

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Unfunded Debt

Unfunded debt consists of:

Savings or Deposit Certificates

Savings Accounts

Postal Life Insurance

General Provident (GP) Fund

Domestic Debt Outstanding

Domestic debt consists of Permanent Debt, Floating Debt, and Unfunded Debt.

The total domestic debt increased over time mainly because floating debt increased significantly.

Changes in one category can affect the overall level of domestic debt.

Trends in Public Debt

Public debt includes debt payable in domestic currency and foreign currency.

Debt levels increased over the years, but debt as a percentage of GDP generally declined.

A declining debt-to-GDP ratio indicates improved debt management and stronger economic performance.

Net public debt is calculated after deducting grants from total public debt.

Budget Deficit of Pakistan

Budget deficit is often measured as a percentage of GDP.

A higher budget deficit increases government borrowing requirements.

Persistent budget deficits contribute to the growth of public debt.

Problems in Measuring Budget Deficit

The traditional budget deficit is calculated as:

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Budget Deficit = Government Spending – Government Revenue

However, this measure may be misleading because it ignores several important factors.

A meaningful budget deficit should:

Adjust for inflation.

Account for government assets.

Include hidden liabilities.

Adjust for business cycle fluctuations.

Inflation and Budget Deficit

Government indebtedness should be measured in real terms rather than nominal terms.

Inflation increases nominal debt even when real debt remains unchanged.

This causes the reported budget deficit to be overstated.

Formula:

$$\Delta D / D = \pi$$

Where:

ΔD = Change in government debt

D = Government debt

π = Inflation rate

Therefore:

$$\Delta D = \pi D$$

The reported budget deficit is overstated by πD .

Example

If:

Inflation Rate = 8.6%

Government Debt = \$495 Billion

Then:

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$$\pi D = 0.086 \times 495 = \$43 \text{ Billion}$$

If the reported budget deficit is \$28 billion:

Actual Budget Balance:

$$\$28 - \$43 = \$15 \text{ Billion Surplus}$$

Capital Assets and Government Debt

A proper assessment of government debt should consider both liabilities and assets.

Government assets should be subtracted from government debt.

Formula:

$$\text{Government Budget Deficit} = \text{Change in Debt} - \text{Change in Assets}$$

This method is known as **Capital Budgeting**.

Capital budgeting recognizes changes in government wealth rather than focusing only on borrowing.

Problems with Capital Budgeting

A major problem is deciding which government expenditures should be classified as capital expenditures.

This makes capital budgeting difficult to implement accurately.

Uncounted Liabilities

Some government liabilities are not included in official budget calculations.

Examples include:

Government employee pensions.

Social security obligations.

Ignoring these liabilities may underestimate the government's future financial burden.

The Business Cycle and Budget Deficit

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Economic fluctuations automatically affect government revenues and expenditures.

During a recession:

Income falls.

Personal tax revenues decrease.

Corporate profits decline.

Corporate tax revenues decrease.

Government welfare spending increases.

As a result, the budget deficit automatically increases.

These changes occur even without any new government policy.

Cyclically Adjusted Budget Deficit

A cyclically adjusted budget deficit removes the effects of economic fluctuations.

It reflects only changes caused by government fiscal policy.

This provides a more accurate measure of government budget performance.

Important Exam Points

Government debt is the accumulation of all past budget deficits.

Domestic debt consists of Permanent Debt, Floating Debt, and Unfunded Debt.

Budget Deficit = Government Spending – Government Revenue.

A meaningful budget deficit should account for inflation, assets, liabilities, and business cycle effects.

Inflation causes the reported budget deficit to be overstated.

The formula for inflation-adjusted debt is $\Delta D = \pi D$.

Capital budgeting considers both government assets and liabilities.

Government Budget Deficit = Change in Debt – Change in Assets.

Uncounted liabilities include pensions and social security obligations.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A recession automatically increases the budget deficit through lower tax revenues and higher government spending.

The cyclically adjusted budget deficit reflects fiscal policy changes only.

Debt-to-GDP ratio is an important indicator of government debt sustainability.

Important MCQs

1. Government debt is defined as:

- A. Current year's government spending
 - B. Current year's tax revenue
 - C. Accumulation of all past budget deficits ✓**
 - D. Total private sector borrowing
-

2. Which of the following is included in Permanent Debt?

- A. Treasury Bills
 - B. Savings Accounts
 - C. Prize Bonds ✓**
 - D. GP Fund
-

3. Inflation causes the reported budget deficit to be:

- A. Understated
 - B. Overstated ✓**
 - C. Unchanged
 - D. Eliminated
-

4. Capital budgeting takes into account:

- A. Only liabilities

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B. Only tax revenues

C. Both assets and liabilities ✓

D. Only government expenditures

5. A cyclically adjusted budget deficit removes the effects of:

A. Inflation

B. Exchange rates

C. Business cycle fluctuations ✓

D. Foreign debt

Short Conceptual Questions

Q1. Why does inflation overstate the budget deficit?

Answer:

Inflation increases nominal government debt even when real debt remains unchanged, causing the reported deficit to appear larger.

Q2. What is a cyclically adjusted budget deficit?

Answer:

It is a budget deficit measure that removes the effects of economic booms and recessions and reflects only fiscal policy changes.

Numerical / Formula Questions

Question 1

Given:

Inflation Rate (π) = 6%

Government Debt (D) = \$800 Billion

Find the amount by which the budget deficit is overstated.

Solution

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Formula:

$$\Delta D = \pi D$$

$$\Delta D = 0.06 \times 800$$

$$\Delta D = 48$$

Answer: Budget Deficit Overstatement = \$48 Billion

Question 2

Given:

Change in Debt = **\$120 Billion**

Change in Assets = **\$40 Billion**

Find the Government Budget Deficit.

Solution

Formula:

Government Budget Deficit = Change in Debt – Change in Assets

$$\text{Government Budget Deficit} = 120 - 40$$

$$\text{Government Budget Deficit} = 80$$

Answer: Government Budget Deficit = \$80 Billion

Lesson 36: Government Debt (Continued) (Summarized Notes)

Traditional View of Government Debt

The traditional view states that a tax cut increases consumer spending and reduces national saving.

Lower national saving raises the interest rate.

Higher interest rates crowd out private investment.

Lower investment reduces capital accumulation.

In the long run, lower capital stock decreases output and economic well-being.

Government Debt in the Solow Growth Model

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The Solow Growth Model explains how government debt affects long-run growth.

Capital accumulation equation:

$$\Delta k = i - \delta k$$

Since investment equals saving multiplied by output:

$$i = sf(k)$$

Therefore:

$$\Delta k = sf(k) - \delta k$$

Lower saving reduces investment.

Lower investment slows capital accumulation.

The economy reaches a lower steady-state level of capital and output.

Too Little Capital and the Golden Rule

If the economy has less capital than the Golden Rule level:

$$K < K_{\text{gold}}^{**}$$

Increasing the saving rate raises future consumption.

Current consumption initially falls.

Future generations benefit from higher output and consumption.

Tax Cut and the IS Curve

Consumption function:

$$C = C(Y - T)$$

A tax cut increases disposable income.

Lower taxes increase consumption.

Higher consumption increases planned expenditure.

Higher expenditure increases equilibrium income.

The IS curve shifts to the right.

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Income increases by the tax multiplier:

$$\Delta Y = \text{MPC} / (1 - \text{MPC}) \times \Delta T$$

International Trade Effects

A tax cut reduces national saving.

Lower national saving causes borrowing from abroad.

Foreign borrowing creates a trade deficit.

The domestic currency appreciates.

Appreciation reduces net exports.

Therefore, international trade weakens the expansionary effect of fiscal policy.

Government Debt in the Mundell-Fleming Model

Goods Market Equation:

$$Y = C(Y - T) + I(r) + G + NX(e)^*$$

Money Market Equation:

$$M/P = L(r, Y)^*$$

Fiscal expansion shifts the IS^* curve to the right.

Results under floating exchange rates:

$$\Delta e > 0$$

$$\Delta Y = 0$$

Exchange rate appreciation offsets the increase in aggregate demand.

Ricardian View of Government Debt

The Ricardian View argues that tax cuts do not increase consumption.

Consumers realize lower taxes today mean higher taxes in the future.

Government borrowing is simply postponed taxation.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Consumers save today's tax cut to pay future taxes.

Current consumption remains unchanged.

Ricardian Equivalence Principle

Government debt is equivalent to future taxes.

Forward-looking consumers treat future taxes like current taxes.

Debt-financed government spending has the same effect as tax-financed spending.

Therefore, government borrowing does not increase aggregate demand.

Consumers and Future Taxes

Consumers consider their lifetime income when making spending decisions.

Future tax liabilities reduce the wealth effect of current tax cuts.

Supporters of the traditional view argue that consumers do not fully consider future taxes.

Therefore, tax cuts may still increase current consumption.

Myopia

Myopia means consumers focus mainly on current income.

They ignore future tax obligations.

A debt-financed tax cut appears to increase their wealth.

As a result, current consumption rises.

This supports the traditional view of government debt.

Borrowing Constraints

Borrowing constraints limit how much consumers can borrow.

Consumers who cannot borrow depend mainly on current income.

A tax cut increases disposable income.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Higher current income increases consumption even if future taxes will rise.

Government borrowing effectively provides consumers with a temporary loan.

Future Generations

According to the traditional view, future taxes will mainly be paid by future generations.

The current generation enjoys higher consumption today.

Future generations bear the burden of government debt.

Debt-financed tax cuts transfer resources from future generations to the present generation.

Important Exam Points

The traditional view states that tax cuts increase consumption and reduce national saving.

Lower saving raises interest rates and crowds out investment.

The Solow Growth Model equation is $\Delta k = sf(k) - \delta k$.

Lower investment reduces long-run capital stock and output.

The Golden Rule level of capital maximizes long-run consumption.

A tax cut shifts the IS curve to the right.

The tax multiplier is $\Delta Y = MPC / (1 - MPC) \times \Delta T$.

Lower national saving causes trade deficits and currency appreciation.

In the Mundell-Fleming Model, fiscal expansion causes $\Delta e > 0$ and $\Delta Y = 0$ under floating exchange rates.

The Ricardian View states that government debt equals future taxes.

Ricardian Equivalence assumes consumers are forward-looking.

Myopia means consumers ignore future taxes and increase current spending.

Borrowing constraints make current income more important than lifetime income.

Debt-financed tax cuts benefit the current generation but burden future generations.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Important MCQs

1. According to the traditional view, a tax cut mainly causes:

- A. Lower consumption
 - B. Higher consumption and lower national saving ✓**
 - C. Lower investment demand
 - D. Lower government spending
-

2. The Solow Growth Model capital accumulation equation is:

- A. $\Delta k = sf(k) - \delta k$ ✓**
 - B. $\Delta k = sf(k) + \delta k$
 - C. $\Delta k = Y - C$
 - D. $\Delta k = G - T$
-

3. According to Ricardian Equivalence, government debt is equivalent to:

- A. Foreign investment
 - B. Money supply
 - C. Future taxes ✓**
 - D. Exports
-

4. Myopia means consumers:

- A. Save all additional income
 - B. Ignore future taxes and focus on current income ✓**
 - C. Borrow unlimited funds
 - D. Reduce consumption after tax cuts
-

5. Under the Mundell-Fleming Model with floating exchange rates, fiscal expansion results in:

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

- A. $\Delta Y > 0$ and $\Delta e = 0$
 - B. $\Delta e > 0$ and $\Delta Y = 0$ ✓
 - C. $\Delta Y < 0$
 - D. No change in exchange rate
-

Short Conceptual Questions

Q1. What is Ricardian Equivalence?

Answer:

Ricardian Equivalence states that government borrowing is equivalent to future taxation, so tax cuts do not change total consumption.

Q2. What are borrowing constraints?

Answer:

Borrowing constraints limit consumers' ability to borrow, making current income more important for consumption decisions.

Numerical / Formula Questions

Question 1

Given:

$$MPC = 0.8$$

$$\text{Tax Cut } (\Delta T) = 100 \text{ billion}$$

Find the increase in income.

Solution

Formula:

$$\Delta Y = MPC / (1 - MPC) \times \Delta T$$

$$\Delta Y = 0.8 / (1 - 0.8) \times 100$$

$$\Delta Y = 0.8 / 0.2 \times 100$$

$$\Delta Y = 4 \times 100$$

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Answer: $\Delta Y = 400$ billion

Question 2

Given:

Saving Function:

$$\Delta k = sf(k) - \delta k$$

Investment = 120 units

Depreciation = 80 units

Find the change in capital stock.

Solution

Formula:

$$\Delta k = \text{Investment} - \text{Depreciation}$$

$$\Delta k = 120 - 80$$

Answer: $\Delta k = 40$ units

Lesson 37: Consumption Theories (Summarized Notes)

John Maynard Keynes and the Consumption Function

The consumption function was a central part of Keynes' theory of economic fluctuations presented in **The General Theory (1936)**.

Keynes proposed three important conjectures.

The **Marginal Propensity to Consume (MPC)** is between 0 and 1.

People consume part of every additional unit of income and save the rest.

The **Average Propensity to Consume (APC)** falls as income rises.

Current income is the primary determinant of consumption.

The interest rate has little effect on consumption.

The Consumption Function

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The Keynesian Consumption Function is:

$$C = \bar{C} + cY$$

Where:

C = Consumption

$\bar{C}$ = Autonomous Consumption

c = Marginal Propensity to Consume (MPC)

Y = Income

The consumption function satisfies Keynes' three conjectures.

The MPC lies between **0 and 1**.

The APC declines as income increases.

Consumption depends mainly on current income.

Average Propensity to Consume (APC)

Formula:

$$APC = C/Y = \bar{C}/Y + c$$

As income increases, $\bar{C}/Y$ decreases.

Therefore, the **Average Propensity to Consume (APC)** falls.

The interest rate is not included in the Keynesian consumption function.

Marginal Propensity to Consume (MPC)

The **Marginal Propensity to Consume (MPC)** measures how much additional consumption results from an additional unit of income.

Formula:

$$MPC = \Delta C / \Delta Y$$

If **MPC = 0.99**, a person spends **99 paisas** out of every extra **1 rupee** earned.

A higher MPC means people spend a larger share of additional income.

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Secular Stagnation and Simon Kuznets

During World War II, economists predicted **secular stagnation**, a long-lasting depression, based on Keynes' consumption function.

The expected depression did not occur after the war.

Simon Kuznets collected long-run data on consumption and income dating back to **1869**.

His research showed that the **Average Propensity to Consume (APC)** remained fairly constant over time.

These findings challenged Keynes' prediction that APC falls as income rises.

The Consumption Puzzle

Household studies and short-run data supported Keynes' consumption theory.

These studies produced the **Short-Run Consumption Function**, where APC falls as income rises.

Long-run data showed that APC remained nearly constant.

This produced the **Long-Run Consumption Function**, where APC does not systematically change with income.

This difference became known as the **Consumption Puzzle**.

Irving Fisher and Intertemporal Choice

Irving Fisher developed the theory of **Intertemporal Choice**.

Consumers make decisions about consuming today versus consuming in the future.

The model explains consumer behavior through:

Consumer budget constraints.

Consumer preferences.

Saving and borrowing decisions.

Consumers choose consumption to maximize satisfaction over different time periods.

Consumer's Budget Constraint

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Assume a consumer lives for two periods.

First-period income = Y_1

Second-period income = Y_2

First-period consumption = C_1

Second-period consumption = C_2

Saving in the first period:

$$S = Y_1 - C_1$$

Second-period consumption:

$$C_2 = (1 + r)S + Y_2$$

Where:

r = Real interest rate

If:

$C_1 < Y_1$, the consumer saves.

$C_1 > Y_1$, the consumer borrows.

Intertemporal Budget Constraint

Combining both periods gives:

$$C_2 = (1 + r)(Y_1 - C_1) + Y_2$$

Rearranging:

$$(1 + r)C_1 + C_2 = (1 + r)Y_1 + Y_2$$

Dividing by $(1 + r)$:

$$C_1 + C_2/(1 + r) = Y_1 + Y_2/(1 + r)$$

This equation is called the **Intertemporal Budget Constraint**.

Importance of the Budget Constraint

If the interest rate is **zero**, total lifetime consumption equals total lifetime income.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

When the interest rate is positive, future income and future consumption are discounted.

The discount factor is:

$$1/(1 + r)$$

Future consumption costs less than current consumption because savings earn interest.

Consumers choose between saving today and borrowing against future income.

Saving and Borrowing Choices

If consumption lies between points **A and B**, consumers save part of their first-period income.

If consumption lies between points **A and C**, consumers borrow to finance current consumption.

Saving increases future consumption.

Borrowing increases present consumption but reduces future consumption.

Important Exam Points

Keynes proposed three consumption conjectures.

The Keynesian Consumption Function is $C = \bar{C} + cY$.

The **Marginal Propensity to Consume (MPC)** lies between **0 and 1**.

The **Average Propensity to Consume (APC)** is $APC = C/Y = \bar{C}/Y + c$.

APC falls as income increases in the Keynesian model.

MPC measures the change in consumption due to a change in income.

Simon Kuznets found that APC remains nearly constant over the long run.

The difference between short-run and long-run APC created the **Consumption Puzzle**.

Irving Fisher developed the theory of **Intertemporal Choice**.

First-period saving is $S = Y_1 - C_1$.

Second-period consumption is $C_2 = (1 + r)S + Y_2$.

The Intertemporal Budget Constraint is:

$$C_1 + C_2/(1 + r) = Y_1 + Y_2/(1 + r).$$

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The discount factor is $1/(1 + r)$.

Consumers save when $C_1 < Y_1$ and borrow when $C_1 > Y_1$.

Important MCQs

1. According to Keynes, the Marginal Propensity to Consume (MPC) lies between:

- A. 1 and 2
 - B. 0 and 1 ✓
 - C. -1 and 0
 - D. Greater than 2
-

2. The Keynesian Consumption Function is:

- A. $C = Y - T$
 - B. $C = \bar{C} + cY$ ✓
 - C. $C = I + G$
 - D. $C = S + T$
-

3. According to Simon Kuznets, the Average Propensity to Consume (APC):

- A. Always decreases rapidly
 - B. Always increases
 - C. Remains fairly constant over the long run ✓
 - D. Depends only on interest rates
-

4. The Intertemporal Budget Constraint is:

- A. $C_1 + C_2 = Y_1 + Y_2$
- B. $C_1 + C_2/(1 + r) = Y_1 + Y_2/(1 + r)$ ✓
- C. $Y_1 = C_1 + S$

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$$D. S = C - Y$$

5. Consumers save when:

A. $C_1 > Y_1$

B. $C_1 < Y_1$ ✓

C. $C_1 = Y_1$

D. Interest rate equals zero

Short Conceptual Questions

Q1. What is the Consumption Puzzle?

Answer:

The Consumption Puzzle is the difference between short-run evidence, where APC falls with income, and long-run evidence, where APC remains nearly constant.

Q2. What is the Intertemporal Budget Constraint?

Answer:

It shows that the present value of lifetime consumption must equal the present value of lifetime income.

Numerical / Formula Questions

Question 1

Given:

Autonomous Consumption ($\bar{C}$) = 200

MPC (c) = 0.8

Income (Y) = 1000

Find Consumption (C).

Solution

Formula:

$$C = \bar{C} + cY$$

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$$C = 200 + (0.8 \times 1000)$$

$$C = 200 + 800$$

Answer: C = 1000

Question 2

Given:

First-period Income (Y_1) = 600

First-period Consumption (C_1) = 450

Real Interest Rate (r) = 10%

Second-period Income (Y_2) = 500

Find Second-period Consumption (C_2).

Solution

Saving:

$$S = Y_1 - C_1$$

$$S = 600 - 450 = 150$$

Formula:

$$C_2 = (1 + r)S + Y_2$$

$$C_2 = (1.10 \times 150) + 500$$

$$C_2 = 165 + 500$$

Answer: $C_2 = 665$

Lesson 38: Consumption Theories (Continued) (Summarized Notes)

Consumer Preferences

Consumer preferences for present and future consumption are represented by **indifference curves**.

An indifference curve shows different combinations of first-period and second-period consumption that provide the same level of satisfaction.

The slope of an indifference curve is called the **Marginal Rate of Substitution (MRS)**.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The **Marginal Rate of Substitution (MRS)** measures how much second-period consumption a consumer is willing to give up to obtain one extra unit of first-period consumption while remaining equally satisfied.

Higher indifference curves represent higher levels of satisfaction.

Consumers always prefer higher indifference curves to lower ones.

Consumer Optimization

Consumers maximize satisfaction by choosing the point where the **budget constraint** touches the highest possible indifference curve.

At the optimum:

The **budget constraint is tangent to the indifference curve.**

At this point, consumers achieve maximum utility with their available income.

Effect of Income on Consumption

An increase in first-period income or second-period income shifts the budget constraint outward.

If consumption in both periods is a **normal good**, higher income increases consumption in both periods.

Consumers enjoy higher lifetime consumption when lifetime income increases.

Effect of Real Interest Rate on Consumption

An increase in the real interest rate affects consumption through two effects.

Income Effect

The income effect occurs because a higher interest rate increases the return on savings.

Consumers may become wealthier and increase consumption.

Substitution Effect

The substitution effect occurs because future consumption becomes relatively cheaper.

Consumers reduce current consumption and increase future consumption.

The total effect depends on which effect is stronger.

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Irving Fisher's Conclusion

Irving Fisher's model shows that the effect of the real interest rate on consumption is uncertain.

Depending on consumer preferences:

Consumption may increase.

Consumption may decrease.

Economic theory alone cannot predict the exact effect.

Savings and the Real Interest Rate

Empirical evidence shows little relationship between savings and the real interest rate.

The income effect and substitution effect usually offset each other.

Therefore, changes in interest rates have only a small effect on total savings.

Borrowing Constraints

A borrowing constraint limits the amount consumers can borrow.

The borrowing constraint is:

$$C_1 \leq Y_1$$

First-period consumption cannot exceed first-period income.

Consumers facing borrowing constraints cannot smooth consumption over time.

Consumption Functions with Borrowing Constraints

Consumers who are **not borrowing constrained** base consumption on their lifetime income.

Consumers who **are borrowing constrained** base consumption mainly on current income.

Therefore, two different consumption functions exist depending on whether borrowing is possible.

High Japanese Saving Rate

Japan has one of the world's highest saving rates.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

High saving contributed to Japan's rapid economic growth after World War II.

According to the Solow Growth Model, higher saving increases investment and raises long-run income.

However, high saving also reduces consumption.

Lower consumption reduces aggregate demand.

According to the IS-LM Model, lower aggregate demand can reduce national income.

Reasons for High Japanese Savings

Households face strict borrowing conditions.

Large down payments are required for housing loans.

The Japanese tax system encourages saving by lightly taxing capital income.

Japanese consumers are generally more patient and more risk-averse.

These factors contribute to Japan's high saving rate.

Important Exam Points

Consumer preferences are represented by **indifference curves**.

The slope of an indifference curve is the **Marginal Rate of Substitution (MRS)**.

Higher indifference curves provide greater satisfaction.

Consumer equilibrium occurs where the **budget constraint is tangent to the highest indifference curve**.

Higher income shifts the budget constraint outward.

Normal goods increase in demand as income rises.

A higher real interest rate creates both **income and substitution effects**.

Irving Fisher concluded that interest rate changes may either increase or decrease consumption.

Savings show little relationship with the real interest rate because income and substitution effects offset each other.

The borrowing constraint is $C_1 \leq Y_1$.

Consumers without borrowing constraints consume according to lifetime income.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Consumers with borrowing constraints consume according to current income.

Japan's high saving rate promotes investment but may reduce aggregate demand.

The Solow Growth Model links higher saving with higher long-run capital and output.

Important MCQs

1. The slope of an indifference curve is called the:

- A. Marginal Product
 - B. Marginal Rate of Substitution (MRS) ✓**
 - C. Average Propensity to Consume
 - D. Budget Multiplier
-

2. Consumer equilibrium occurs where:

- A. Income equals saving
 - B. Consumption equals income
 - C. Budget constraint is tangent to the highest indifference curve ✓**
 - D. Interest rate is zero
-

3. The borrowing constraint is expressed as:

- A. $C_1 \geq Y_1$
 - B. $C_1 \leq Y_1$ ✓**
 - C. $S = C - Y$
 - D. $Y = C + I$
-

4. According to Irving Fisher, the effect of a higher real interest rate on consumption is:

- A. Always positive
- B. Always negative

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C. Uncertain because income and substitution effects work in opposite directions ✓

D. Equal to zero

5. One reason for Japan's high saving rate is:

A. High inflation

B. Low investment

C. Strict borrowing conditions and high down payments ✓

D. High unemployment

Short Conceptual Questions

Q1. What is the Marginal Rate of Substitution (MRS)?

Answer:

The Marginal Rate of Substitution measures how much future consumption a consumer is willing to exchange for one additional unit of current consumption while maintaining the same satisfaction.

Q2. What is a borrowing constraint?

Answer:

A borrowing constraint limits consumers from borrowing beyond their current income, so first-period consumption cannot exceed first-period income.

Numerical / Formula Questions

Question 1

Given:

First-period Income (Y_1) = 800

First-period Consumption (C_1) = 600

Find Saving (S).

Solution

Formula:

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$$S = Y_1 - C_1$$

$$S = 800 - 600$$

Answer: $S = 200$

Question 2

Given:

First-period Income (Y_1) = 1000

Second-period Income (Y_2) = 1200

Real Interest Rate (r) = 20%

Find the present value of lifetime income.

Solution

Formula:

$$\text{Present Value} = Y_1 + Y_2/(1 + r)$$

$$\text{Present Value} = 1000 + 1200/1.20$$

$$\text{Present Value} = 1000 + 1000$$

Answer: Present Value of Lifetime Income = 2000

Lesson 39: Consumption Theories (Continued) (Summarized Notes)

John Maynard Keynes and the Consumption Function

Keynes proposed that consumption depends mainly on current income.

The consumption function has three important properties.

The **Marginal Propensity to Consume (MPC)** lies between 0 and 1.

The **Average Propensity to Consume (APC)** falls as income rises.

Current income is the primary determinant of consumption, while the interest rate has little effect.

Simon Kuznets and the Consumption Puzzle

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Simon Kuznets found that the Average Propensity to Consume remained nearly constant over long periods.

This contradicted Keynes' prediction that APC decreases as income rises.

The puzzle was why Keynes' theory worked well for short-run household data but failed for long-run data.

Irving Fisher and Intertemporal Choice

Irving Fisher developed the theory explaining how rational consumers allocate consumption between the present and the future.

Consumers consider:

Their available resources.

Their preferences.

Their budget constraints.

Consumers maximize satisfaction by deciding how much to consume today and how much to save for future consumption.

The **Intertemporal Budget Constraint** determines total lifetime consumption.

Franco Modigliani and the Life-Cycle Hypothesis

Franco Modigliani developed the **Life-Cycle Hypothesis** to explain consumption behavior over a person's lifetime.

People save during their working years and spend those savings after retirement.

Consumers attempt to maintain a stable level of consumption throughout life.

Life-Cycle Consumption Function

Lifetime resources consist of current wealth and future earnings.

Formula:

$$C = (W + RY) / T$$

Or,

$$C = (1/T)W + (R/T)Y$$

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Where:

C = Consumption

W = Wealth

Y = Annual Income

R = Remaining working years

T = Total expected lifetime

Example:

If **T = 50** and **R = 30**

C = 0.02W + 0.6Y

Aggregate Consumption Function:

C = α W + β Y

Where:

α = MPC out of Wealth

β = MPC out of Income

Solving the Consumption Puzzle

According to the Life-Cycle Hypothesis,

APC = C/Y = α (W/Y) + β

In the short run, wealth and income do not increase proportionally, so APC changes.

In the long run, wealth and income grow together, making **W/Y** nearly constant.

Therefore, APC remains stable over time.

This explains the consumption puzzle identified by Kuznets.

Consumption, Income, and Wealth over the Life Cycle

People earn income during working years.

They save part of their income while working.

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After retirement, income falls but consumption remains relatively stable by using accumulated savings.

Consumption remains smoother than income throughout life.

Consumption and Saving of Elderly

Research shows elderly people save more than predicted by the Life-Cycle Hypothesis.

Reasons include:

Precautionary saving for uncertain medical expenses or longer life expectancy.

Desire to leave inheritances (bequests) to children.

These factors reduce the speed at which wealth is spent after retirement.

Milton Friedman and the Permanent-Income Hypothesis

Milton Friedman proposed that consumption mainly depends on **Permanent Income** rather than current income.

Current income consists of two parts:

$$Y = Y^P + Y^T$$

Where:

Y^P = Permanent Income

Y^T = Transitory Income

Permanent income is expected to continue in the future.

Transitory income is temporary and unexpected.

Consumers smooth consumption by saving or borrowing when temporary income changes occur.

Consumption Function:

$$C = \alpha Y^P$$

Average Propensity to Consume:

$$APC = \alpha Y^P / Y$$

If $Y > Y^P$, APC falls.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

If $Y < Y^p$, APC rises.

Robert Hall and the Random-Walk Hypothesis

Robert Hall combined the Permanent-Income Hypothesis with Rational Expectations.

He concluded that changes in consumption are largely unpredictable.

Consumption follows a **Random Walk**, meaning consumers adjust spending only when unexpected information changes their permanent income.

Future consumption cannot be accurately predicted from past consumption alone.

Important Exam Points

Keynes proposed that **MPC lies between 0 and 1**.

Keynes believed **APC decreases as income rises**.

Kuznets found that **APC remains nearly constant in the long run**.

Irving Fisher introduced the **Intertemporal Choice Model**.

The Life-Cycle Hypothesis was developed by **Franco Modigliani**.

People save during working years and dissave after retirement.

Life-Cycle Consumption Function is $C = (W + RY)/T$.

Aggregate Consumption Function is $C = \alpha W + \beta Y$.

The Life-Cycle Hypothesis explains the **Consumption Puzzle**.

Permanent-Income Hypothesis was developed by **Milton Friedman**.

Current income equals **Permanent Income + Transitory Income** ($Y = Y^p + Y^t$).

Consumption mainly depends on **Permanent Income**.

Robert Hall proposed the **Random-Walk Hypothesis**.

Random Walk states that changes in consumption are unpredictable.

Important MCQs

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

1. Who proposed the Life-Cycle Hypothesis?

- A. John Maynard Keynes
 - B. Milton Friedman
 - C. Franco Modigliani ✓**
 - D. Robert Hall
-

2. According to the Permanent-Income Hypothesis, consumption mainly depends on:

- A. Current income
 - B. Government spending
 - C. Permanent income ✓**
 - D. Tax revenue
-

3. Current income is divided into:

- A. Wealth and Savings
 - B. Investment and Consumption
 - C. Permanent Income and Transitory Income ✓**
 - D. Taxes and Expenditure
-

4. According to Robert Hall, consumption follows a:

- A. Business cycle
 - B. Consumption function
 - C. Random Walk ✓**
 - D. Phillips Curve
-

5. The Aggregate Consumption Function under the Life-Cycle Hypothesis is:

- A. $C = \bar{C} + cY$

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B. $C = \alpha Y$

C. $C = \alpha W + \beta Y$ ✓

D. $C = Y - T$

Short Conceptual Questions

Q1. What is Permanent Income?

Answer:

Permanent income is the portion of income that people expect to continue over a long period and use to determine their consumption.

Q2. What is the Random-Walk Hypothesis?

Answer:

The Random-Walk Hypothesis states that changes in consumption are unpredictable because consumers adjust spending only when new unexpected information becomes available.

Numerical / Formula Questions

Question 1

Given:

Wealth (W) = Rs. 500,000

Annual Income (Y) = Rs. 300,000

Remaining Working Years (R) = 30

Total Lifetime (T) = 50

Find Consumption (C).

Solution

Formula:

$$C = (W + RY)/T$$

$$C = (500,000 + 30 \times 300,000) / 50$$

$$C = (500,000 + 9,000,000) / 50$$

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$$C = 9,500,000 / 50$$

Answer: C = Rs. 190,000

Question 2

Given:

Permanent Income (Y^p) = Rs. 800,000

$$\alpha = 0.8$$

Find Consumption.

Solution

Formula:

$$C = \alpha Y^p$$

$$C = 0.8 \times 800,000$$

Answer: C = Rs. 640,000

Lesson 40: Investment Theories (Summarized Notes)

Investment and Economic Fluctuations

Investment is the most volatile component of GDP.

During recessions, investment spending falls sharply, causing a decline in overall economic output.

The IS-LM model assumes that investment depends negatively on the real interest rate.

Investment Function:

$$I = I(r)$$

An increase in the real interest rate reduces investment spending.

Investment theories explain why investment changes during booms and recessions.

Three Types of Investment Spending

Investment spending is classified into three major types.

Business Fixed Investment.

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Residential Investment.

Inventory Investment.

Investment models help explain:

Why investment is negatively related to the interest rate.

What causes the investment function to shift.

Why investment rises during economic expansions and falls during recessions.

Business Fixed Investment

Business Fixed Investment accounts for nearly **three-fourths ($\frac{3}{4}$)** of total investment spending.

It includes purchases of machinery, factories, computers, vehicles, office equipment, and other productive assets.

The **Neoclassical Model of Investment** explains business fixed investment by comparing the benefits and costs of owning capital.

The three major factors affecting investment are:

The Marginal Product of Capital (MPK).

The Interest Rate.

Tax Rules.

Production Firms and Rental Firms

The Neoclassical Model assumes two types of firms.

Production Firms produce goods and services using rented capital.

Rental Firms purchase capital goods and rent them to production firms.

In reality, most firms perform both production and investment functions.

The Rental Price of Capital

A production firm rents capital until the benefit from an additional unit of capital equals its cost.

The benefit of capital is measured by the **Marginal Product of Capital (MPK)**.

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The cost is the **Real Rental Price of Capital (R/P)**.

Equilibrium Condition:

$$MPK = R/P$$

The MPK curve slopes downward because each additional unit of capital produces less additional output.

The supply of capital is fixed in the short run.

The rental price adjusts to balance the demand and supply of capital.

Cobb-Douglas Production Function

The Cobb-Douglas Production Function describes how labor and capital produce output.

Formula:

$$Y = AK^{\alpha}L^{1-\alpha}$$

Where:

Y = Output

A = Technology

K = Capital

L = Labor

α = Capital's share of output

Marginal Product of Capital (MPK)

The Marginal Product of Capital for the Cobb-Douglas Production Function is:

$$MPK = \alpha A(L/K)^{1-\alpha}$$

Since the rental price equals MPK,

$$R/P = \alpha A(L/K)^{1-\alpha}$$

The real rental price increases when:

Capital stock decreases.

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Labor employment increases.

Technology improves.

The Cost of Capital

Rental firms earn income by renting capital but must bear ownership costs.

The three components of capital cost are:

Interest Cost.

Capital Gain or Loss.

Depreciation.

Total Cost of Capital:

$$\text{Total Cost} = iPK - \Delta PK + \delta PK$$

Or,

$$\text{Total Cost} = PK(i - \Delta PK/PK + \delta)$$

Where:

PK = Purchase price of capital

i = Interest rate

ΔPK = Change in capital price

δ = Depreciation rate

The cost of capital depends on interest rates, depreciation, and changes in capital prices.

Example of Cost of Capital

Suppose:

Purchase Price of Car (**PK**) = **Rs.1,000,000**

Interest Rate (**i**) = **10%**

Capital Gain = **6%**

Depreciation = **20%**

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Interest Cost:

100,000

Capital Gain:

60,000

Depreciation Cost:

200,000

Total Cost:

$100,000 - 60,000 + 200,000 = \text{Rs.}240,000$

Important Exam Points

Investment is the most volatile component of GDP.

Investment Function is **$I = I(r)$** .

Higher real interest rates reduce investment.

Business Fixed Investment accounts for about **$\frac{3}{4}$ of total investment**.

The Neoclassical Model explains business investment decisions.

The three determinants of investment are **MPK, Interest Rate, and Tax Rules**.

Production firms rent capital, while rental firms own capital.

The equilibrium condition is **$MPK = R/P$** .

The Cobb-Douglas Production Function is **$Y = AK^\alpha L^{1-\alpha}$** .

The Marginal Product of Capital is **$MPK = \alpha A(L/K)^{1-\alpha}$** .

The Real Rental Price equals the Marginal Product of Capital.

The Cost of Capital includes interest, depreciation, and capital gains or losses.

Total Cost of Capital is **$iPK - \Delta PK + \delta PK$** .

Improved technology increases the rental price of capital.

Higher labor employment increases the marginal product of capital.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Important MCQs

1. Investment is considered the most volatile component of:

- A. Consumption
 - B. Government Spending
 - C. GDP ✓
 - D. Exports
-

2. The Neoclassical Model mainly explains:

- A. Consumer behavior
 - B. Inflation
 - C. Business Fixed Investment ✓
 - D. Money supply
-

3. A production firm rents capital until:

- A. Interest rate becomes zero
 - B. Output equals consumption
 - C. $MPK = R/P$ ✓
 - D. Investment becomes zero
-

4. The Cobb-Douglas Production Function is:

- A. $C = \bar{C} + cY$
 - B. $Y = C + I + G$
 - C. $Y = AK^\alpha L^{1-\alpha}$ ✓
 - D. $MV = PY$
-

5. The total cost of capital is:

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A. $iPK + \Delta PK$

B. $iPK - \delta PK$

C. $iPK - \Delta PK + \delta PK$ ✓

D. $PK - \delta$

Short Conceptual Questions

Q1. What is Business Fixed Investment?

Answer:

Business Fixed Investment is spending by firms on long-term productive assets such as machinery, factories, vehicles, and equipment.

Q2. Why does MPK decrease as capital increases?

Answer:

Due to diminishing returns, each additional unit of capital adds less to total output than the previous unit.

Numerical / Formula Questions

Question 1

Given:

Purchase Price of Capital (PK) = Rs. 800,000

Interest Rate (i) = 12%

Capital Gain = Rs. 40,000

Depreciation = Rs. 120,000

Find the Total Cost of Capital.

Solution

Formula:

Total Cost = $iPK - \Delta PK + \delta PK$

Interest Cost = $12\% \times 800,000 = \text{Rs. } 96,000$

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

$$\text{Total Cost} = 96,000 - 40,000 + 120,000$$

$$\text{Total Cost} = \text{Rs.176,000}$$

Answer: Total Cost of Capital = Rs.176,000

Question 2

Given:

$$\alpha = 0.3$$

$$A = 2$$

$$\text{Labor (L)} = 100$$

$$\text{Capital (K)} = 25$$

Find the Marginal Product of Capital.

Solution

Formula:

$$\text{MPK} = \alpha A (L/K)^{1-\alpha}$$

$$\text{MPK} = 0.3 \times 2 \times (100/25)^{0.7}$$

$$\text{MPK} = 0.6 \times 4^{0.7}$$

$$4^{0.7} \approx 2.64$$

$$\text{MPK} \approx 0.6 \times 2.64$$

Answer: MPK $\approx$ 1.58

Lesson 41: Investment Theories (Continued) (Summarized Notes)

The Cost of Capital

The cost of capital represents the total cost of purchasing and owning capital goods.

Total Cost of Capital:

$$\text{Cost of Capital} = iPK - \Delta PK + \delta PK$$

Or,

$$\text{Cost of Capital} = PK(i - \Delta PK/PK + \delta)$$

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Where:

PK = Price of Capital

i = Nominal Interest Rate

ΔPK = Change in Capital Price

δ = Depreciation Rate

The cost of capital depends on the price of capital, interest rate, depreciation, and changes in capital prices.

Assuming capital prices rise with the overall inflation rate:

$$\Delta PK/PK = \pi$$

Since,

$$r = i - \pi$$

The Cost of Capital becomes:

$$\text{Cost of Capital} = PK(r + \delta)$$

Real Cost of Capital

The Real Cost of Capital measures the cost of owning capital relative to the economy's output.

Formula:

$$\text{Real Cost of Capital} = (PK/P)(r + \delta)$$

Where:

PK/P = Relative Price of Capital

r = Real Interest Rate

δ = Depreciation Rate

An increase in the real interest rate increases the real cost of capital.

Determinants of Investment

Rental firms compare the revenue earned from renting capital with the cost of owning it.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Real Profit per unit of capital is:

$$\text{Profit Rate} = \text{Revenue} - \text{Cost}$$

$$\text{Profit Rate} = R/P - (PK/P)(r + \delta)$$

Since:

$$MPK = R/P$$

Profit becomes:

$$\text{Profit Rate} = MPK - (PK/P)(r + \delta)$$

Investment depends on the difference between the Marginal Product of Capital and the Cost of Capital.

If $MPK > \text{Cost of Capital}$, firms increase investment.

If $MPK < \text{Cost of Capital}$, firms reduce investment.

Net Investment Equation:

$$\Delta K = \text{In}[MPK - (PK/P)(r + \delta)]$$

The Investment Function

Total Business Fixed Investment consists of Net Investment plus Replacement Investment.

Investment Function:

$$I = \text{In}[MPK - (PK/P)(r + \delta)] + \delta K$$

A lower real interest rate reduces the cost of capital and increases investment.

A higher real interest rate raises the cost of capital and reduces investment.

Therefore, the investment function slopes downward.

Technological improvements increase MPK and shift the investment function upward.

Long-Run Equilibrium

Over time, firms adjust their capital stock.

If $MPK > \text{Cost of Capital}$, capital stock increases and MPK gradually falls.

If $MPK < \text{Cost of Capital}$, capital stock decreases and MPK gradually rises.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

In the long run, equilibrium occurs when:

$$MPK = (PK/P)(r + \delta)$$

At this point, firms have no incentive to increase or decrease their capital stock.

Taxes and Investment

Government tax policies strongly influence investment decisions.

Two important tax provisions are:

Corporate Income Tax.

Investment Tax Credit.

Corporate Income Tax affects investment depending on how taxable profits are calculated.

If profits exceed the cost of capital, firms continue investing.

Investment Condition:

$$R/P > (PK/P)(r + \delta)$$

Treatment of Depreciation

Economic theory measures depreciation using the current value of capital.

Tax laws usually calculate depreciation using the historical purchase cost.

This difference affects firms' taxable profits and investment incentives.

Investment Tax Credit

The Investment Tax Credit encourages firms to purchase new capital goods.

It reduces taxes based on investment spending.

The credit lowers the effective purchase price of capital.

A lower capital cost increases investment and shifts the investment function upward.

Swedish Investment Funds System

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Sweden used investment subsidies to stabilize aggregate demand.

During economic slowdowns, the government offered temporary investment subsidies to encourage investment.

During economic recoveries, these subsidies were withdrawn.

Eventually, investment subsidies became a permanent feature of Sweden's tax policy.

Important Exam Points

The Cost of Capital is $PK(r + \delta)$.

The Real Cost of Capital is $(PK/P)(r + \delta)$.

Investment depends on the difference between **MPK** and the **Cost of Capital**.

Profit Rate is $MPK - (PK/P)(r + \delta)$.

Net Investment is $\Delta K = \ln[MPK - (PK/P)(r + \delta)]$.

The Investment Function is $I = \ln[MPK - (PK/P)(r + \delta)] + \delta K$.

A lower real interest rate increases investment.

A higher real interest rate reduces investment.

Technological improvement shifts the investment function upward.

Long-run equilibrium occurs when $MPK = (PK/P)(r + \delta)$.

Corporate Income Tax affects firms' investment incentives.

Investment Tax Credit lowers the effective cost of capital.

Sweden used investment subsidies to stabilize aggregate demand.

Important MCQs

1. The real cost of capital is:

A. $PK(r - \delta)$

B. $MPK + \delta$

C. $(PK/P)(r + \delta)$ ✓

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D. PK/P

2. Firms increase investment when:

- A. $MPK < \text{Cost of Capital}$
 - B. Interest rate rises
 - C. $MPK > \text{Cost of Capital}$ ✓
 - D. Depreciation becomes zero
-

3. The Investment Function is:

- A. $C = \bar{C} + cY$
 - B. $Y = C + I + G$
 - C. $I = \ln[MPK - (PK/P)(r + \delta)] + \delta K$ ✓
 - D. $MV = PY$
-

4. The Investment Tax Credit encourages firms by:

- A. Increasing depreciation
 - B. Raising interest rates
 - C. Reducing the effective purchase price of capital ✓
 - D. Increasing corporate taxes
-

5. Long-run equilibrium occurs when:

- A. $MPK = 0$
 - B. Investment = 0
 - C. $MPK = (PK/P)(r + \delta)$ ✓
 - D. Interest rate = Inflation
-

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Short Conceptual Questions

Q1. Why does a lower real interest rate increase investment?

Answer:

A lower real interest rate reduces the cost of capital, making investment more profitable and encouraging firms to purchase more capital goods.

Q2. What is the purpose of the Investment Tax Credit?

Answer:

The Investment Tax Credit reduces the effective cost of purchasing capital goods, encouraging firms to increase investment.

Numerical / Formula Questions

Question 1

Given:

$$PK = \text{Rs.}500,000$$

$$P = 2$$

$$\text{Real Interest Rate } (r) = 8\%$$

$$\text{Depreciation Rate } (\delta) = 12\%$$

Find the Real Cost of Capital.

Solution

Formula:

$$\text{Real Cost of Capital} = (PK/P)(r + \delta)$$

$$PK/P = 500,000 \div 2 = 250,000$$

$$r + \delta = 0.08 + 0.12 = 0.20$$

$$\text{Real Cost} = 250,000 \times 0.20$$

Answer: Real Cost of Capital = Rs.50,000

Question 2

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Given:

$$MPK = 18$$

$$(PK/P)(r + \delta) = 12$$

Find the Profit Rate.

Solution

Formula:

$$\text{Profit Rate} = MPK - (PK/P)(r + \delta)$$

$$\text{Profit Rate} = 18 - 12$$

Answer: Profit Rate = 6

Lesson 42: Investment Theories (Continued) (Summarized Notes)

The Stock Market and Tobin's q

The stock market is the market where corporate ownership shares (stocks) are bought and sold.

James Tobin introduced **Tobin's q** to explain firms' investment decisions.

Tobin's q Formula:

$$q = \text{Market Value of Installed Capital} / \text{Replacement Cost of Installed Capital}$$

Where:

Market Value of Installed Capital = Value determined by the stock market.

Replacement Cost of Installed Capital = Current cost of purchasing the same capital.

If $q > 1$, firms increase investment because market value exceeds replacement cost.

If $q < 1$, firms reduce investment because replacing capital is not profitable.

A higher marginal product of capital increases profitability, raises stock prices, and increases Tobin's q.

The Stock Market as an Economic Indicator

Stock market movements often provide signals about future economic activity.

A significant fall in stock prices may indicate an upcoming recession.

A decline in stock prices lowers Tobin's q, reducing firms' incentives to invest.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Lower stock prices also reduce household wealth, decreasing consumption and aggregate demand.

Falling stock prices may reflect weaker technological progress and slower economic growth.

Financing Constraints

Firms finance investment through financial markets.

Common sources of finance include:

Bank loans.

Corporate bonds.

Selling shares in the stock market.

The neoclassical model assumes firms can always obtain funds if they are willing to pay the cost of capital.

However, some firms face **financing constraints**, limiting their ability to raise external funds.

During recessions, lower profits reduce internal funds available for investment.

Firms with easy access to financial markets continue investing despite temporary recessions.

Firms facing financing constraints often reduce investment even when profitable opportunities exist.

Residential Investment

Residential investment includes spending on newly constructed houses.

It covers houses purchased by both homeowners and landlords.

For simplicity, the model assumes all houses are owner-occupied.

Residential investment is explained using two connected markets:

The market for existing houses determines the equilibrium housing price.

The housing price determines the amount of new residential construction.

Housing Market Equilibrium

The relative price of housing adjusts until housing demand equals housing supply.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Construction firms purchase labor and materials to build houses.

Their production costs depend on the general price level (P).

Their revenues depend on the housing price (PH).

A higher housing price increases builders' profits and encourages more residential construction.

Higher relative housing prices lead to greater residential investment.

Residential Investment and Tobin's q

Residential investment follows logic similar to Tobin's q theory.

Business investment depends on the market value of installed capital relative to replacement cost.

Residential investment depends on the relative price of housing.

Higher expected housing returns increase housing demand, raise house prices, and encourage new construction.

Expected future rental income plays an important role in determining housing demand.

Important Exam Points

Tobin's q measures the ratio of market value of installed capital to its replacement cost.

Formula: $q = \text{Market Value of Installed Capital} / \text{Replacement Cost of Installed Capital}$.

If $q > 1$, firms increase investment.

If $q < 1$, firms reduce investment.

Higher profitability raises Tobin's q and stimulates investment.

Stock market declines often signal future economic slowdowns.

Lower stock prices reduce wealth, consumption, and aggregate demand.

Financing constraints limit firms' ability to invest despite profitable opportunities.

Residential investment includes the construction of new housing.

Housing prices determine the level of residential investment.

Higher housing prices encourage more residential construction.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Residential investment follows principles similar to Tobin's q theory.

Expected future rental income increases housing demand and investment.

Important MCQs

1. Tobin's q is defined as:

- A. GDP divided by investment
 - B. Consumption divided by income
 - C. Market value of installed capital divided by replacement cost of installed capital ✓**
 - D. Capital divided by labor
-

2. Firms are most likely to increase investment when:

- A. $q < 1$
 - B. $q > 1$ ✓**
 - C. $q = 0$
 - D. $q = 1/2$
-

3. A major decline in stock prices usually indicates:

- A. Higher inflation
 - B. Possible future recession ✓**
 - C. Higher exports
 - D. Lower unemployment
-

4. Financing constraints refer to:

- A. High production costs
- B. Low labor productivity
- C. Limited ability of firms to raise funds for investment ✓**

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D. Government budget deficits

5. Residential investment mainly depends on:

A. Exchange rate

B. Government debt

C. Relative price of housing ✓

D. Population only

Short Conceptual Questions

Q1. What is Tobin's q?

Answer:

Tobin's q is the ratio of the market value of installed capital to its replacement cost and helps determine firms' investment decisions.

Q2. What are financing constraints?

Answer:

Financing constraints are limits that prevent firms from raising enough funds for profitable investment projects.

Numerical / Formula Questions

Question 1

Given:

Market Value of Installed Capital = **Rs. 120 million**

Replacement Cost of Installed Capital = **Rs. 100 million**

Find **Tobin's q**.

Solution

Formula:

$q = \text{Market Value of Installed Capital} / \text{Replacement Cost of Installed Capital}$

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

$$q = 120 / 100$$

Answer: $q = 1.2$

Since $q > 1$, firms are encouraged to increase investment.

Question 2

Given:

Market Value of Installed Capital = **Rs. 90 million**

Replacement Cost of Installed Capital = **Rs. 120 million**

Find **Tobin's q** and state the investment decision.

Solution

Formula:

$$q = \text{Market Value of Installed Capital} / \text{Replacement Cost of Installed Capital}$$

$$q = 90 / 120$$

$$q = 0.75$$

Answer: $q = 0.75$

Since $q < 1$, firms are likely to reduce investment.

Lesson 43: Investment Theories (Continued) (Summarized Notes)

Inventory Investment

Inventory investment refers to goods that businesses keep in storage for future sale or production.

Although inventory investment is one of the smallest components of GDP, it is highly volatile and plays an important role during economic fluctuations.

During recessions, firms stop replacing inventories as goods are sold, causing inventory investment to become negative.

The main reasons firms hold inventories are:

Production smoothing.

Improving production efficiency.

Stock-out avoidance.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Work in process.

Reasons for Holding Inventories

Production Smoothing

When sales are high, firms may produce less than they sell and meet excess demand by using existing inventories.

This helps maintain stable production levels over time.

Inventories as a Factor of Production

Holding inventories allows firms to operate more efficiently.

Raw materials, supplies, and intermediate goods help maintain continuous production.

Stock-Out Avoidance

Firms keep inventories to avoid running out of goods when customer demand suddenly increases.

This helps prevent lost sales and dissatisfied customers.

Work in Process

Partially completed products are also counted as inventory.

These unfinished goods are called **work in process**.

Seasonal Fluctuations and Production Smoothing

Many economists expected firms to use inventories to smooth production during seasonal demand changes.

However, evidence shows that firms generally adjust production according to seasonal sales instead of relying heavily on inventories.

For example, fan manufacturers usually increase production during high-demand seasons rather than producing large inventories in advance.

Thus, seasonal production patterns closely follow seasonal sales patterns.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The Accelerator Model of Inventories

The Accelerator Model assumes firms hold inventories proportional to their level of output.

When output increases, firms require more raw materials, supplies, and finished goods.

When output decreases, firms reduce inventory holdings.

The desired inventory stock is:

$$N = \beta Y$$

Where:

N = Inventory stock

Y = Output

β = Desired inventory-output ratio

Inventory investment equals the change in inventory stock.

Inventory Investment Formula:

$$I = \Delta N = \beta \Delta Y$$

Inventory investment depends on changes in output rather than the level of output.

Predictions of the Accelerator Model

When output rises, firms increase inventories, resulting in positive inventory investment.

When output falls, firms reduce inventories, resulting in negative inventory investment.

The model suggests inventory investment depends on whether the economy is expanding or contracting.

Inventories and the Real Interest Rate

Inventory investment is also influenced by the real interest rate.

Holding inventories involves an opportunity cost because firms could earn interest by selling goods immediately.

Higher real interest rates increase the cost of holding inventories.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

As a result, firms reduce inventory holdings when interest rates rise.

Therefore, higher real interest rates decrease inventory investment.

Important Exam Points

Inventory investment refers to goods kept in storage for future production or sale.

Inventory investment becomes negative during recessions.

The four main reasons for holding inventories are production smoothing, production efficiency, stock-out avoidance, and work in process.

Production smoothing uses inventories to meet temporary increases in sales.

Work in process refers to partially completed goods.

The Accelerator Model assumes inventories are proportional to output.

Desired Inventory Formula: $N = \beta Y$.

Inventory Investment Formula: $I = \Delta N = \beta \Delta Y$.

Inventory investment depends on changes in output.

Higher output increases inventory investment.

Higher real interest rates reduce inventory investment because they increase the opportunity cost of holding inventories.

Important MCQs

1. Inventory investment becomes negative when:

- A. Output increases
 - B. Inflation falls
 - C. Firms reduce inventories during recessions ✓**
 - D. Interest rates become zero
-

2. According to the Accelerator Model, desired inventories are:

- A. Equal to interest rates

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B. Independent of output

C. Proportional to output ✓

D. Equal to government spending

3. The Accelerator Model is represented by:

A. $I = C + G$

B. $N = \beta Y$ ✓

C. $Y = C + I$

D. $MPK = R/P$

4. Work in process refers to:

A. Finished goods only

B. Raw materials only

C. Partially completed products ✓

D. Imported goods

5. An increase in the real interest rate will generally:

A. Increase inventory investment

B. Increase inventory demand

C. Reduce inventory investment ✓

D. Have no effect on inventories

Short Conceptual Questions

Q1. Why do firms hold inventories?

Answer:

Firms hold inventories to smooth production, improve efficiency, avoid stock shortages, and maintain work in process.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Q2. What does the Accelerator Model explain?

Answer:

The Accelerator Model explains that inventory investment changes according to changes in output.

Numerical / Formula Questions

Question 1

Given:

$$\beta = 0.25$$

Increase in Output (ΔY) = **Rs. 400 million**

Find the inventory investment.

Solution

Formula:

$$I = \beta \Delta Y$$

$$I = 0.25 \times 400$$

$$I = \text{Rs. 100 million}$$

Answer: Inventory Investment = Rs. 100 million

Question 2

Given:

Desired inventory ratio (β) = **0.30**

Output (Y) = **Rs. 800 million**

Find the desired inventory stock.

Solution

Formula:

$$N = \beta Y$$

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

$$N = 0.30 \times 800$$

$$N = \text{Rs. 240 million}$$

Answer: Desired Inventory Stock = Rs. 240 million

Lesson 44: Money & Banking – Money Supply (Summarized Notes)

Money Supply

Money supply refers to the total amount of money available in the economy.

Earlier, money supply was explained simply as the currency held by the public.

In reality, money supply is determined not only by the Central Bank but also by households and commercial banks.

Money supply includes both currency and demand deposits used for daily transactions.

Money Supply Formula:

$$M = C + D$$

Where:

M = Money Supply

C = Currency held by the public

D = Demand Deposits

Determinants of Money Supply

The Central Bank influences money supply through monetary policy and open market operations.

Households determine how much money they hold as currency or deposit in banks.

Commercial banks affect money supply by accepting deposits and creating loans.

Therefore, money supply depends on the interaction of the Central Bank, households, and banks.

100% Reserve Banking

In a 100% Reserve Banking System, banks accept deposits but do not lend any money.

All deposited funds are kept as reserves.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Reserves are deposits that banks receive but do not lend.

Some reserves are held in bank vaults, while most are kept at the Central Bank.

Since banks do not make loans, they earn only small service fees.

The banking system does not create additional money.

Balance Sheet of a 100% Reserve Bank

If the public deposits **\$1,000** into a bank, the bank's balance sheet is:

Assets

Reserves = **\$1,000**

Liabilities

Deposits = **\$1,000**

Because every deposited dollar is held in reserve, the total money supply remains unchanged.

Money Supply before banking = **\$1,000**

Money Supply after banking = **\$1,000**

Thus, **100% reserve banking does not change the money supply.**

Fractional Reserve Banking

In a Fractional Reserve Banking System, banks keep only a fraction of deposits as reserves.

The remaining deposits are used to make loans to households and businesses.

Banks earn profits by charging interest on these loans.

Banks maintain enough reserves to meet normal withdrawal demands.

As long as deposits approximately equal withdrawals, banks do not need to keep all deposits as reserves.

Reserve-Deposit Ratio

The **Reserve-Deposit Ratio** is the fraction of deposits that banks keep as reserves.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Reserve-Deposit Ratio Formula:

Reserve-Deposit Ratio = Reserves / Deposits

Banks may also hold **Excess Reserves**, which are reserves above the required minimum.

Money Creation under Fractional Reserve Banking

Unlike 100% reserve banking, fractional reserve banking allows banks to create money.

When banks lend part of their deposits, borrowers spend the money, which is redeposited into the banking system.

These new deposits allow further lending.

As this process continues, the total money supply increases.

Thus, commercial banks create money through the lending process.

Important Exam Points

Money supply consists of currency and demand deposits.

Money Supply Formula: $M = C + D$.

Money supply is determined by the Central Bank, households, and commercial banks.

100% reserve banking keeps all deposits as reserves.

100% reserve banking does not create new money.

Fractional reserve banking keeps only part of deposits as reserves.

Banks create money by making loans.

The Reserve-Deposit Ratio equals reserves divided by deposits.

Excess reserves are reserves above the required minimum.

Fractional reserve banking increases the money supply.

Important MCQs

1. The money supply is represented by:

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A. $M = C - D$

B. $M = C + D$ ✓

C. $M = C \times D$

D. $M = D - C$

2. In a 100% reserve banking system, banks:

A. Lend all deposits

B. Create new money

C. Keep all deposits as reserves ✓

D. Hold no reserves

3. Fractional reserve banking allows banks to:

A. Eliminate deposits

B. Reduce currency

C. Create money through lending ✓

D. Eliminate reserves

4. The Reserve-Deposit Ratio is defined as:

A. $\text{Deposits} \div \text{Loans}$

B. $\text{Currency} \div \text{Deposits}$

C. $\text{Reserves} \div \text{Deposits}$ ✓

D. $\text{Loans} \div \text{Reserves}$

5. Excess reserves are:

A. Deposits above loans

B. Currency outside banks

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C. Reserves above the required minimum ✓

D. Government reserves only

Short Conceptual Questions

Q1. Why does 100% reserve banking not increase the money supply?

Answer:

Because banks keep all deposits as reserves and do not make loans, no new money is created.

Q2. How do commercial banks create money?

Answer:

Commercial banks create money by lending a portion of deposited funds under the fractional reserve banking system.

Numerical / Formula Questions

Question 1

Given:

Currency held by the public (C) = **Rs. 700 million**

Demand Deposits (D) = **Rs. 500 million**

Find the money supply.

Solution

Formula:

$$M = C + D$$

$$M = 700 + 500$$

$$M = \text{Rs. 1,200 million}$$

Answer: Money Supply = Rs. 1,200 million

Question 2

Given:

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Reserves = **Rs. 150 million**

Deposits = **Rs. 600 million**

Find the Reserve-Deposit Ratio.

Solution

Formula:

Reserve-Deposit Ratio = Reserves / Deposits

Reserve-Deposit Ratio = $150 / 600$

Reserve-Deposit Ratio = **0.25 = 25%**

Answer: Reserve-Deposit Ratio = 25%

Lesson 45: Money & Banking (Continued) (Summarized Notes)

How the Central Bank Controls the Money Supply

The central bank controls the money supply through three main policy tools.

These are:

Open Market Operations

Reserve Requirements

Discount Rate

Open Market Operations

Open market operations involve the purchase and sale of government securities by the central bank.

When the central bank **buys government bonds**, it injects money into the economy, increasing the monetary base and money supply.

When the central bank **sells government bonds**, money is withdrawn from circulation, reducing the monetary base and money supply.

Reserve Requirements

Reserve requirements specify the minimum fraction of deposits that banks must keep as reserves.

An increase in the reserve requirement raises the reserve-deposit ratio.

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A higher reserve ratio lowers the money multiplier and decreases the money supply.

Discount Rate

The discount rate is the interest rate charged by the central bank on loans to commercial banks.

A lower discount rate encourages banks to borrow more reserves.

More bank borrowing increases the monetary base and expands the money supply.

A higher discount rate discourages borrowing and contracts the money supply.

Limitations of Central Bank Control

Although the central bank has powerful tools, it cannot perfectly control the money supply.

Banks may hold **excess reserves** beyond legal requirements.

Banks may also borrow freely from the discount window.

These actions can change the money supply differently from what the central bank expects.

Money Demand

Money demand refers to the amount of money people wish to hold for transactions and other purposes.

Two major theories explain money demand:

Classical Theory

Keynesian Theory

Classical Theory of Money Demand

The Quantity Theory of Money states that demand for real money balances is proportional to income.

Formula:

$$(M/P)^d = kY$$

Where:

$(M/P)^d$ = Real money demand

ECO403 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

k = Constant proportion

Y = Income

Higher income increases the demand for real money balances.

Keynesian Theory of Money Demand

Keynes proposed that money demand depends on both income and the interest rate.

Formula:

$$(M/P)^d = L(i, Y)$$

Where:

i = Interest rate

Y = Income

Higher income increases money demand.

Higher interest rates reduce money demand because holding money becomes more costly.

Functions of Money

Money performs three important functions.

Unit of Account

Store of Value

Medium of Exchange

The last two functions mainly explain why people demand money.

Portfolio Theory of Money Demand

Portfolio theories view money as one asset among many in an individual's wealth portfolio.

Money demand depends on the returns offered by alternative assets and total wealth.

Money Demand Function:

$$(M/P)^d = L(r_s, r_b, \pi_e, W)$$

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Where:

r_s = Expected return on stocks

r_b = Expected return on bonds

π_e = Expected inflation

W = Real wealth

Higher returns on stocks or bonds reduce money demand.

Higher expected inflation also lowers money demand.

Greater wealth increases money demand.

Measures of Money Supply

Different measures include different financial assets.

C: Currency only

M1: Currency + Demand Deposits + Travelers' Checks + Other Checkable Deposits

M2: M1 + Savings Deposits + Small Time Deposits + Money Market Accounts

M3: M2 + Large Time Deposits + Repurchase Agreements + Institutional Money Market Funds

Portfolio theories better explain broader measures like **M2** than narrow measures like **M1**.

Transactions Theory of Money Demand

Transactions theories emphasize money as a medium of exchange.

People hold money because it makes buying goods and services easier.

Holding money has an opportunity cost because it earns little or no interest.

Baumol-Tobin Model of Cash Management

The Baumol-Tobin model explains the optimal amount of cash people should hold.

Holding more cash reduces trips to the bank.

Holding more cash also increases forgone interest.

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People balance these costs and benefits to determine optimal cash balances.

More frequent bank visits reduce average cash holdings.

Money Creation

In a fractional reserve banking system, banks create money by making loans.

Each bank keeps only a fraction of deposits as reserves and lends the remainder.

Money Creation Formula:

Total Money Supply = $(1/rr) \times \text{Initial Deposit}$

Where:

rr = Reserve-Deposit Ratio

Example:

Initial Deposit = **\$1000**

Reserve Ratio = **20% (0.2)**

Money Supply = **$(1/0.2) \times 1000 = \$5000$**

Financial Intermediation

Financial intermediation is the process through which financial institutions transfer funds from savers to borrowers.

Banks collect savings from households and provide loans to businesses and consumers for investment.

Model of Money Supply

Three important exogenous variables determine the money supply.

Monetary Base (B)

Reserve-Deposit Ratio (rr)

Currency-Deposit Ratio (cr)

Definitions:

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$$M = C + D$$

$$B = C + R$$

Money Multiplier Formula:

$$m = (cr + 1)/(cr + rr)$$

Money Supply:

$$M = m \times B$$

Because every increase in the monetary base creates a multiplied increase in money supply, the monetary base is called **High-Powered Money**.

Money Multiplier Example

Given:

Monetary Base (B) = \$500 billion

Reserve Ratio (rr) = 0.1

Currency Ratio (cr) = 0.6

Money Multiplier:

$$m = (0.6 + 1)/(0.6 + 0.1)$$

$$m = 1.6/0.7 = 2.3$$

Money Supply:

$$M = 2.3 \times 500 = \$1150 \text{ billion}$$

Factors Affecting Money Supply

An increase in the monetary base increases money supply.

A lower reserve-deposit ratio increases the money multiplier.

A lower currency-deposit ratio leaves more reserves in banks and increases money creation.

Important Exam Points

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The central bank controls money supply through **Open Market Operations, Reserve Requirements, and Discount Rate.**

Buying government bonds increases the monetary base and money supply.

Selling government bonds decreases the money supply.

Higher reserve requirements reduce the money multiplier.

A lower discount rate encourages banks to borrow more reserves.

The Classical money demand function is $(M/P)^d = kY$.

The Keynesian money demand function is $(M/P)^d = L(i, Y)$.

Money performs three functions: **Unit of Account, Store of Value, and Medium of Exchange.**

Portfolio theory relates money demand to wealth and expected returns on assets.

Baumol-Tobin Model explains the optimal cash balance by balancing convenience and forgone interest.

Fractional reserve banking creates money through repeated lending.

Money Multiplier Formula:

$$m = (cr + 1)/(cr + rr)$$

Money Supply Formula:

$$M = m \times B$$

The monetary base is also called **High-Powered Money.**

Important MCQs

1. Which policy tool increases the money supply when the central bank buys government bonds?

A. Reserve Requirement

B. Open Market Operations ✓

C. Discount Rate Increase

D. Tax Policy

2. According to the Keynesian theory, money demand depends on:

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- A. Inflation only
 - B. GDP only
 - C. Interest rate and income ✓**
 - D. Government spending
-

3. The money multiplier is calculated as:

- A. $rr + cr$
 - B. rr/cr
 - C. $(cr + 1)/(cr + rr)$ ✓**
 - D. B/M
-

4. High-powered money refers to:

- A. Currency only
 - B. Bank deposits
 - C. Monetary Base ✓**
 - D. Government bonds
-

5. In the Baumol-Tobin model, holding more cash mainly results in:

- A. Higher stock prices
 - B. Higher opportunity cost due to forgone interest ✓**
 - C. Lower income
 - D. Higher inflation
-

Short Conceptual Questions

Q1. Why is the monetary base called high-powered money?

Answer:

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The monetary base is called high-powered money because each unit of it creates multiple units of money through the banking system.

Q2. How does a lower reserve-deposit ratio affect money supply?

Answer:

A lower reserve ratio allows banks to lend more money, increasing the money multiplier and expanding the money supply.

Numerical / Formula Questions

Question 1

Given:

Monetary Base (B) = \$600 billion

Reserve Ratio (rr) = 0.2

Currency Ratio (cr) = 0.3

Find the money multiplier and money supply.

Solution

Formula:

$$m = (cr + 1)/(cr + rr)$$

$$m = (0.3 + 1)/(0.3 + 0.2)$$

$$m = 1.3/0.5$$

$$m = 2.6$$

Money Supply:

$$M = m \times B$$

$$M = 2.6 \times 600$$

Answer: M = \$1560 billion

Question 2

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Suppose:

Initial Deposit = \$2,000

Reserve Ratio (rr) = 10% (0.10)

Find the maximum money supply created.

Solution

Formula:

Money Supply = $(1/rr) \times \text{Initial Deposit}$

Money Supply = $(1/0.10) \times 2000$

Money Supply = 10×2000

Answer: Money Supply = \$20,000

THE END

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